Optimal Market Making in Intraday Electricity Markets via Hawkes Process Modeling

by

Robert Hennings

Submitted to the

Chair of Financial Economics at the

Institute for Quantitative Business and Economic Research



Faculty of Business, Economics and Social Science

Masterthesis

Master of Science in Quantitative Finance

Christian-Albrechts-University of Kiel

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Submitted: XX.09.2026

Abstract

DEEP Reinforcement (RL) is one of the most successful and recent techniques in the field of Artificial Intelligence (AI). It has been successfully applied to various fields, including robotics, gaming, and finance. In this paper, we explore the application of Deep RL in the context of algorithmic trading in the electricity market. We propose a novel approach that combines Deep RL with advanced neural network architectures to optimize trading strategies. Our method leverages historical market data and real-time information to make informed trading decisions. We evaluate our approach using a comprehensive dataset from the electricity market and compare it with traditional trading strategies. The results demonstrate that our Deep RL-based method outperforms conventional approaches, achieving higher returns and lower risk. This research contributes to the growing body of literature on AI applications in finance and provides valuable insights for practitioners in the energy trading sector.

Keywords: **Reinforcement Learning**, **Deep Learning**, **Electricity Market**, **Trading**, **Algorithmic Trading**, **Energy Trading**, **Financial Economics**, **Quantitative Finance**, **Machine Learning**, **Artificial Intelligence**, **Neural Networks**, **Time Series Analysis**, **Forecasting**, **Decision Making**, **Optimization**, **Stochastic Optimal Control**, **Market Making**, **Market Microstructure**,

JEL Classification: **C61**, **C63**, **G11**, **G17**, **Q41**

Acknowledgements

As a fairly storable commodity, data is often cited as the new oil, therefore I would like to express my sincere gratitude to Leon von Essen, ETD Product Designer at the EUREX, for providing me with access to the EPEX SPOT data, without this project would not have been possible. I am truly thankful for his support and collaboration, even though my internship dates back almost four years. As he is, in fact, a *market maker*, him supporting research devoted to this topic is a well rounded circle. May the volume be with him.

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List of Abbreviations

RES	Renewable Energy Sources
CID	Continuous Intraday
IDM	Intraday Market
ICE	Intercontinental Exchange
EEX	European Energy Exchange
EPEX SPOT	European Power Exchange Spot Market
IDA	Intraday Auction
LOB	Limit Order Book
API	Application Programming Interface
MO	Market Order
LO	Limit Order
CA	Cancellations
HJB	Hamilton-Jacobi-Bellman Equation
DRL	Deep Reinforcement Learning
MLE	Maximum Likelihood Estimation
DA	Day-Ahead Market
GWh	Gigawatt Hour
CET	Central European Time
TSO	Transmission System Operator
SDAC	Single Day-Ahead Coupling
NEMO	Nominated Electricity Market Operator
PPA	Power Purchase Agreement
MTU	Market Time Unit
SIDC	Single Intraday Coupling
SOB	Shared Orderbook
MWh	Megawatt Hour
FOK	Fill or Kill
IOC	Immediate or Cancel
AON	All or None
MM	Market Maker
ZI	Zero Intelligence
DMM	Designated Market Maker
i.i.d	independent and identically distributed
HPP	Homogeneous Poisson Process
NHPP	Non-Homogeneous Poisson Process
MHP	Marked Hawkes Process
SDHP	State Dependent Hawkes Process
SDMHP	State Dependent Marked Hawkes Process
NHP	Neural Hawkes Process
KKT	Karush-Kuhn-Tucker
LSTM	Long Short-Term Memory
L{X}	(Price) Level X of the LOB

DPP	Dynamic Programming Principle
PDE	Partial Differential Equation
ACF	Autocorrelation Function
PACF	Partial Autocorrelation Function
PDF	Probability Density Function
CDF	Cumulative Probability Density Function
ECDF	Empirical Cumulative Probability Density Function
QQ	Quantile-Quantile
MLE	Maximum Likelihood Estimation
EEG	Erneuerbare Energien Gesetz
EUPHEMIA	Pan-European Hybrid Electricity Market Integration Algorithm

List of Symbols

t	discrete time index t
T	discrete terminal time index T
$\lambda(t)$	Hawkes Process intensity
δ	order book depth
Δp	tick size
$P(\delta)$	fill-probability of order book depth
V	order volume
q	inventory level
$\mathcal{A}$	Set of Admissible Strategies
ϕ	(Running) Inventory penalty
λ	(Conditional) Intensity
$\mathcal{F}$	Filtration
S	Midprice
X	Cash-Process
W	Brownian Motion

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1 Introduction

The ever increasing power generation attributed to renewable energy sources (RES), predominantly in Germany within Europe, with a current share of 57,2 % of produced gross-electricity by RES (2025)[1], highlights the rapid decarbonization of the energy sector.[2] By regulatory design, RES have a feed-in priority[3], meaning that they are guaranteed to be integrated into the electricity grid and are given priority over other sources of electricity generation, thereby directly impacting the price formation process.

As the RES production, as just one of many influencial external variables, is itself heavily weather dependent, this uncertainty is transferred into the (european) electricity markets, resulting in increased price volatility and a growing demand for balancing the short term imbalances as well as forecasting errors in the continuous intraday markets (CID), where the financial trading of these positions is carried out.

1.1 The financialization of the electricity market

In Europe, the main exchanges for trading electricity are the Intercontinental Exchange (ICE), the NordPool, the European Energy Exchange (EEX) and the European Power Exchange (EPEX SPOT). The latter one, acting as the reference spot market, where the short term trading of electricity takes place, what this thesis focuses on.

These exchanges provide a platform for trading electricity contracts, including futures, options, and spot contracts. The spot trading is organized in a CID market and three intraday auctions (IDA), where the CID allows for trading of electricity contracts with delivery within the next 24 hours, while the IDA is a single auction that takes place at a specific times each day, allowing for trading of electricity contracts with delivery on the following day. The CID market offers 1-hour contracts, 30-Minute contracts, and the most granular 15-Minute contracts, covering the entire 24-hour period of the next following day. The IDA, on the other hand, focuses on 1-hour contracts for the next day, providing a platform for market participants to adjust their positions based on updated information and forecasts. More details about the market structure and trading mechanics of the EPEX SPOT will be provided in chapter 2.1 The structure of the European Power Market.

The trading in these contracts is facilitated through a limit order book (LOB), where buyers and sellers submit their orders, which are then matched based on price and time priority. The LOB provides a transparent view of the market, allowing participants to see the available liquidity and make informed trading decisions. This *order-driven market* structure is a key feature of the electricity markets, as it allows for efficient price discovery and facilitates the trading of electricity contracts in a competitive environment, in

[1] See Data from [Arb26], last accessed: 2026-05-13 19:08.
[2] [See Eur26, p.2]
[3] See [Eur18] for the relevant EU directive and [Eur24] for the support scheme recommendation, as well as the national EEG [Deu23].

contrast to a *quote-driven market* structure, where market makers provide liquidity by quoting buy and sell prices for the contracts. The dynamics of the LOB are influenced by various factors, including the arrival of limit orders and market orders, cancellations, and modifications of existing orders. In order for a smooth operating and liquid market, it is crucial to have a sufficient number of market participants, including so called market makers, who provide liquidity by continuously quoting buy and sell prices for the contracts, making sure incoming orders can be executed efficiently at all times. Liquidity provision is carried out by these market makers at a rapid speed pocketing the bid-ask spread, which is the difference between the highest price a buyer is willing to pay and the lowest price a seller is willing to accept. As the bid-ask spread is usually only a few cents wide, at best, earning this gap has to be done several thousands of times a day, which is only possible through the use of algorithmic trading strategies, which are often based on statistical and machine learning techniques, designed to quickly analyze huge amounts of market data, identify trading opportunities, and execute trades at high speeds. With an average daily order volume of 1,242,762 orders in the EPEX SPOT CID market in 2019, this leaves no real other options, as to make use of algorithmic trading strategies, to be able to compete in this market.[4] This process is essential for maintaining an efficient and liquid market, as it allows for quick execution of trades and helps to reduce price volatility. The presence of market makers also encourages more participants to enter the market, as they can be assured of liquidity and competitive pricing. In recent years, the role of market makers has become increasingly important in the electricity markets, with many participants often using algorithmic trading strategies, what is also evident in the share of submitted orders to the EPEX SPOT via Application Programming Interfaces (APIs), that is around 65% and represents around 47% of traded volume.[5]

The discrepancy between automatically and human interacted manual trading volume is due that the automatically submitted orders are often smaller in size, but more frequent, while the manually submitted orders are often larger in size, but less frequent, what later motivates modelling decisisons with respect to variable order sizes.

This financialization highlights the economic importance of efficient trading strategies and risk management techniques in the electricity markets, as market participants seek to optimize their returns while managing the risks associated with price volatility and uncertainty. The unique characteristics of the electricity markets, such as the non-storability of electricity and the influence of weather conditions on supply and demand, further emphasize the need for sophisticated modeling approaches that can capture these dynamics effectively. In this context, the application of Hawkes processes for optimal market making in intraday electricity markets offers a promising avenue for developing advanced trading strategies that can adapt to the complex and dynamic nature of these markets, what this thesis essentially covers.

Here the main overall approach resembles around the idea that the LOB can be treated

[4] 2019 figures, see [EPE19] for more details.

[5] 2019 figures, see [EPE19] for more details, as of 20-05-2026 455 Exchange Members are registered and 4XX APIs.

as stochastic system, also referred to as *zero-intelligence model* but will be enhanced by including some priors like non-constant order intensities or dependency structures [6], in contrast to the often used second main approach, treating the LOB as a strategic game between market participants, where the actions of one participant can influence the actions of others, leading to a complex interplay of strategies and counter-strategies, also known as *perfect rational agent-based model*. By modeling the LOB as a stochastic queueing system, we can also allow for flexible modelling techniques for the involved order types, that can be submitted **market orders** (MO), **Limit Orders** (LO) and **cancellations**, which are the main drivers of the LOB dynamics and are explained in more detail in 2.2.1 Order Types.[7]

Increasing trading volumes, Increasing renewables energy production, increasing uncertainty through renewables through weather induced volatility flexible planning and balancing through intraday market mechanism, cite volume increase, connection to financial markets and LOB, efficient price discovery, high share of algorithmic trading unique stylized facts that need to be modelled,

1.2 Main Research Objectives of the Thesis

The main objective of this thesis is to explore the application of Hawkes processes for optimal market making in intraday electricity markets. Specifically, we aim to develop a modeling framework that captures the dynamics of the limit order book (LOB) and the interactions between different order types, such as limit orders, market orders, and cancellations. By leveraging the self-exciting nature of Hawkes processes, we seek to model the arrival of orders and their impact on the LOB, allowing us to develop trading strategies that can adapt to the evolving market conditions. Through empirical analysis and numerical simulations, we aim to evaluate the performance of our proposed approach and compare it with traditional market making strategies, ultimately contributing to the advancement of trading techniques in the context of intraday electricity markets. Taking into account the often not properly considered dynamics of Cross excitation, time varying baseline intensities and general intensities, external excitement through updated forecasts and external variables, we seek to develop a comprehensive modeling framework that can effectively capture the complex dynamics of the LOB and provide insights into optimal market making strategies in the context of intraday electricity markets. The main research questions that this thesis seeks to address are:

- How can Hawkes processes be effectively applied to model the dynamics of the limit order book in intraday electricity markets?
- What are the key stylized facts of the limit order book in the EPEX SPOT CID market, and how can they be incorporated into the modeling framework?

[6] See [Jai+24a].
[7] See [Gou+13].

- How does the performance of the proposed Hawkes process-based market making strategy compare to traditional market making strategies in terms of profitability and risk management?
- What are the implications of the findings for practitioners in the energy trading sector, and how can they be used to inform trading strategies and risk management techniques in intraday electricity markets?

In answering the above objectives, the main contributions are

This thesis is aimed at obtaining the optimal, dynamic quote distances from the best bid/ask using stochastic optimal control theory solving the Hamilton-Jacobi-Bellman (HJB) equation. This is done, proposing different, extending modelling frameworks, all making use of the multivariate marked Hawkes model with time-varying baseline intensities. The bid and ask side will be modelled separately, allowing for interactions, as well as per side, then the three main components, affecting price building, namely Market Order Arrivals, Limit Order Arrivals and Cancellations.By starting from this baseline model framework with six (interaction) channels, it will be extended to the main model by also incorporating the first three price levels per side. Two additional extensions will be reviewed, namely the *state-dependent* and the *marked* one. In the state-dependent extension, the intensities will be dependent on parameters like imbalance, spread, queue size, inventory, time to delivery and time of day. In the marked extension, order size and price impact will be used as marks, especially for MO and LO. A possible optional extension could be the use of Deep Reinforcement Learning (DRL), since especially in the state-dependent extensions, we face a rich state space, which can be hard to solve through traditional methods, but is well suited for DRL, which can be used to learn optimal policies. For the estimation strategy, the maximum likelihood estimation (MLE) will be the main workhorse, but we will consider also non-parametric spline based estimation for the excitation kernel functions.

1.3 Structure of the Thesis

The thesis is organised into three main parts, where the first one comprises the theoretical background, the second one the empirical characteristics of the LOB in the EPEX SPOT CID market and the third one the mathematical modelling and its numerical analysis. The structure of the thesis is outlined as follows, after having laid out the motivation for the chosen research topic along with its main research objectives and their added new insights in 1 Introduction the theoretical background follows. In chapter 2 Theoretical Background we provide the necessary theoretical details on the structure of the european power market, its organization, functional mechanics and the key problem of market making, what this thesis focuses on. Following in 3 Empirical LOB Characteristics we present the empirical characteristics, or *stylized facts*, of the LOB in the EPEX SPOT CID market, which act as the main orientation basis and benchmark for the modelling

approach in the later chapters.[8] Being equipped with the needed background knowledge about the market mechanics and typical patterns, the central modelling aspcets are covered in 4 Mathematical Modelling. There the Hawkes process, along with its underlying mathematical theory, is introduced and the modelling approach for optimal market making is discussed. As the proposed modelling is put to test in chapter 5 Numerical Analysis and Simulation being used on actual, empirical data from the EPEX SPOT CID market, where the performance of the optimal market making framework is evaluated, the discussion is carried out in chapter 6 Discussion. Final conclusions are drawn in chapter 7 Conclusion, where the main findings of the thesis are summarized, their implications are discussed, and avenues for future research are suggested.

[8] See [Jai+24a].

2 Theoretical Background

We begin by introducing the structure of the European Power Market, with a particular focus on the EPEX SPOT market, which serves as the primary context for our study. We then delve into the organization of trading which is based on the LOB system, where the main order-variants and the matching mechanism will be discussed. The chapter will be closed by analysing the key problem of market making, which is the main focus of this work.

2.1 The structure of the European Power Market

The core structure of the european wholesale power market is fourfold, consisting of the Derivatives Market, the Day-Ahead or Forward Market, the Intraday Market, and the Balancing Market.[9] Their (chronological) order can be seen in Figure 2.1, where a special emphasis is put on the timely ordering of the different markets and their events, as this plays a crucial role for modelling. Their order follows the time-to-delivery of the traded contracts, with the Derivatives Market being the furthest away from delivery, followed by the Day-Ahead Market, the Intraday Market, and finally the Balancing Market, which is closest to delivery. In the following, these components will be briefly discussed in their

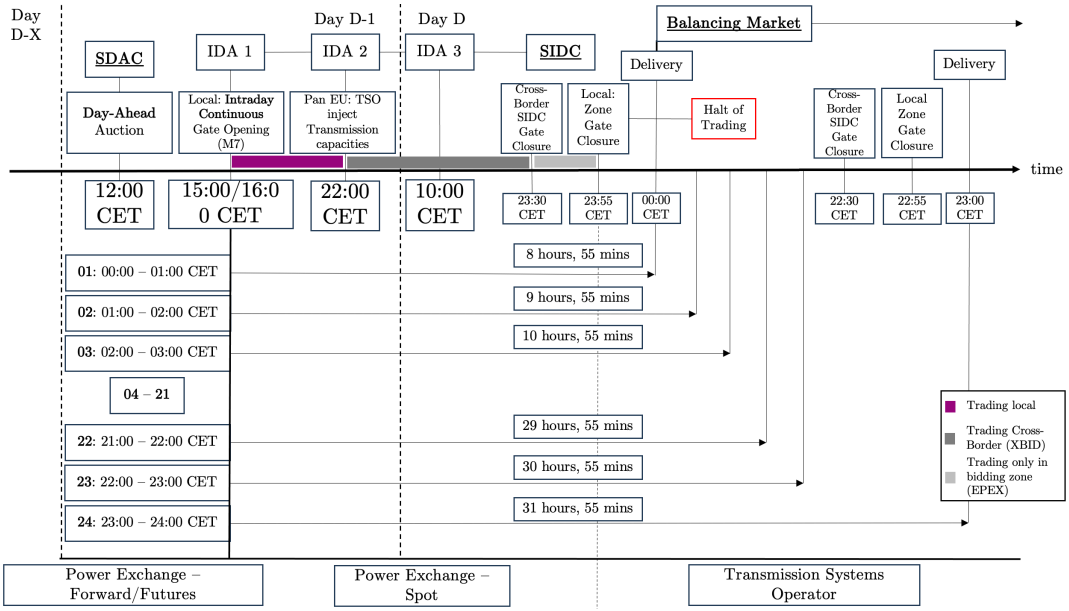


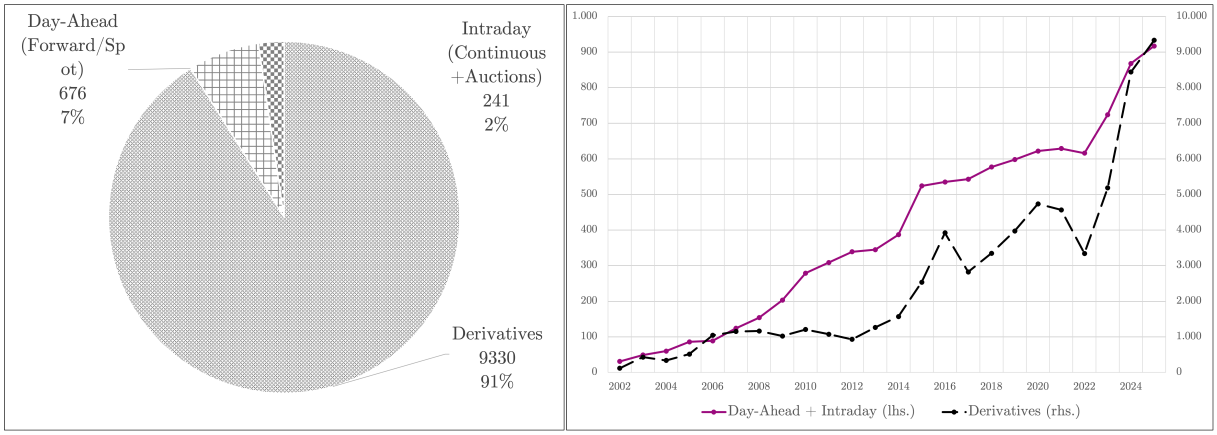
Figure 2.1: Schematic Overview of the European Power Market Design with exemplary hourly products in chronological order.[10]

chronological order, that also resembles the markets importance based on the Trading

[9] The Day-Ahead and the Intraday Market together are most commonly referred to as *Spot Markets*, [see Bun25, p.136.]

[10] Based on [Kie19], as of 16.05.2026.

Volume, that can be seen in Figure 2.2. In the left panel of Figure 2.2, we can see the



Panel a): Annual Trading Volume in GWh of the year 2025, split into the core structural sub-markets. Panel b): Historic development of the trading volume for the two categories Derivatives Market and Spot Market, where the latter one includes the Day-Ahead and Intraday Market volumes.

Figure 2.2: European Power Market Trading Volume: Historic (panel b, right) and Year 2025 (panel a, left) Figures.[11]

annual Trading Volume in GWh of the year 2025, split into the core structural sub-markets, where the clear dominance of the Derivatives Market is evident, where around 91% of traded volumes is concentrated. In the right panel of Figure 2.2, we can see the historic development of the trading volume for the two categories Derivatives Market and Spot Market, where the latter one includes the Day-Ahead and Intraday Market volumes. From the historic development, the nearly 10:1 relation of the volumes is verified.

2.1.1 The Derivatives Market

The derivatives market of the EEX is the most one in terms of trading volume, where the traded contracts are primarily settled financially[12], meaning that the contracts are not necessarily linked to a physical delivery of electricity, but rather serve as financial instruments for hedging and speculation over the medium term.[13] The product range includes futures, options, and other derivatives based on electricity prices, that can be traded for various delivery periods, such as respective next 11 days and the next 2 weekends, the current and the next 4 weeks, the current and the next 24 months, 11 quarters and 10 years. Exchange trading can be conducted from 8:00 am to 6:00 pm (CET). In the european landscape, alongside the EEX, also the International Commodities Exchange (ICE) and the Nasdaq OMX are competing for trading volumes for a broader range of delivery areas.

[11] Based on [Kie19], as of 16.05.2026.
[12] [See Bun25, p. 136]
[13] A full list of all tradable instruments and their specifications can be found here.

2.1.2 The Day-Ahead Market

In contrast to the Derivatives Market of the EEX, where the traded contracts are primarily settled financially, a physical delivery is mandatory on the Day-Ahead Market (also referred to as *Spot Market*).[14] The Day-Ahead Market is a sealed-bid, uniform price, multi-unit implicit auction.[15] Its main purpose is to already lock in certain volumes for the next day (D). Based on newly arrived forecasts, updated information and the need to balance out the supply and demand, market participants can adjust their positions in the Intraday Market. Market participants can submit so-called bidding curves, stating precisely how much power they would want to generate or consume at each price tick between the minimum and maximum prices at each operational subperiod of the next following day, i.e. 24 hours, 30 Minutes and 15 Minutes, resulting in 96 subperiods in total. Submitted prices at which the produces would want to sell power is by regulatory design set to the marginal cost of production, that dictates a natural merit-order hierarchy of the different production technologies. RES naturally operate at the lowest marginal costs, often assumed to be zero, therefore occupying the lowest price levels and the left side of the supply curve, while conventional power plants, such as coal, gas and nuclear, operate at higher marginal costs, occupying the higher price levels and the right side of the supply curve. These biddingcurves have to be submitted up until the gate closure time, which is typically around 12:00 CET on the day before delivery (D-1), allowing them to react to the latest information. Most general Market areas are further divided into bidding zones, which are defined by the transmission system operators (TSOs) and represent areas with similar supply and demand conditions. The geographical area of Germany is divided into four control areas, according to the four TSOs, which are TenneT (DE1), 50Hertz (DE2), Amprion (DE3) and TransnetBW (DE4), that are encoded in the send orders as can be seen in the data. The price formation is then carried out through a uniform price auction, where the market clearing price is determined by the intersection of the aggregated supply and demand curves, which are formed by the submitted orders (for every granularity of the next following day, i.e. 96 15-Min subperiods). The market clearing price is the price at which the total quantity of electricity demanded equals the total quantity supplied, and all transactions are executed at this price. It also serves as a reference price for the Intraday Market and the Balancing Market, as it reflects the market's expectations for the next day. All power producers that have submitted orders at or below the market clearing price will be scheduled to produce electricity, and are awarded the market clearing price for their production. An exemplary supply and demand curve for the market area DE-LU can be seen in Figure 2.3, for the delivery date of the 28th of May, 00:00-00:15. To enhance european coupling, optimizing cross border flow and ultimately welfare, the Day-Ahead Market is coupled across the different bidding zones (Single Day Ahead Coupling - SDAC), meaning that the market clearing price is determined simultaneously across all bidding zones[16], taking into account the cross-border transmission capacities

[14] [See Bun25, p. 136]
[15] [See Bal22, p.233]
[16] Exact pricing is carried out by the EUPHEMIA algorithm.

and the supply and demand conditions in each zone, which is a crucial aspect for the price formation process and the resulting price dynamics in the Intraday Market, as the latter one is directly linked to the Day-Ahead Market through the trading of the same contracts.

The exchanges landscape in Europe is here more fragmented, as there are multiple smaller nominated electricity market operators (NEMOs) that operate Day-Ahead Markets.

Microstructurally: batch market: aggregates liquidity in a single point in time, it aggregates liquidity at a single point in time, eliminating intra-auction adverse selection and any latency race (cf. Budish et al., 2015). Its uniform price is the reference for the entire downstream complex: futures settle against it, intraday prices are quoted (in the empirical literature) as spreads to it, and many support schemes and PPAs index to it. Order types include simple hourly (now quarter-hourly) step and interpolated orders, block orders (fill-or-kill across multiple MTUs, with linked and exclusive variants), and, on some NEMOs, more exotic constructs (loop blocks, big blocks, scalable complex orders); a storage order type is under public consultation with expected SDAC go-live in Q4 2027 (ENTSO-E, 2026b).

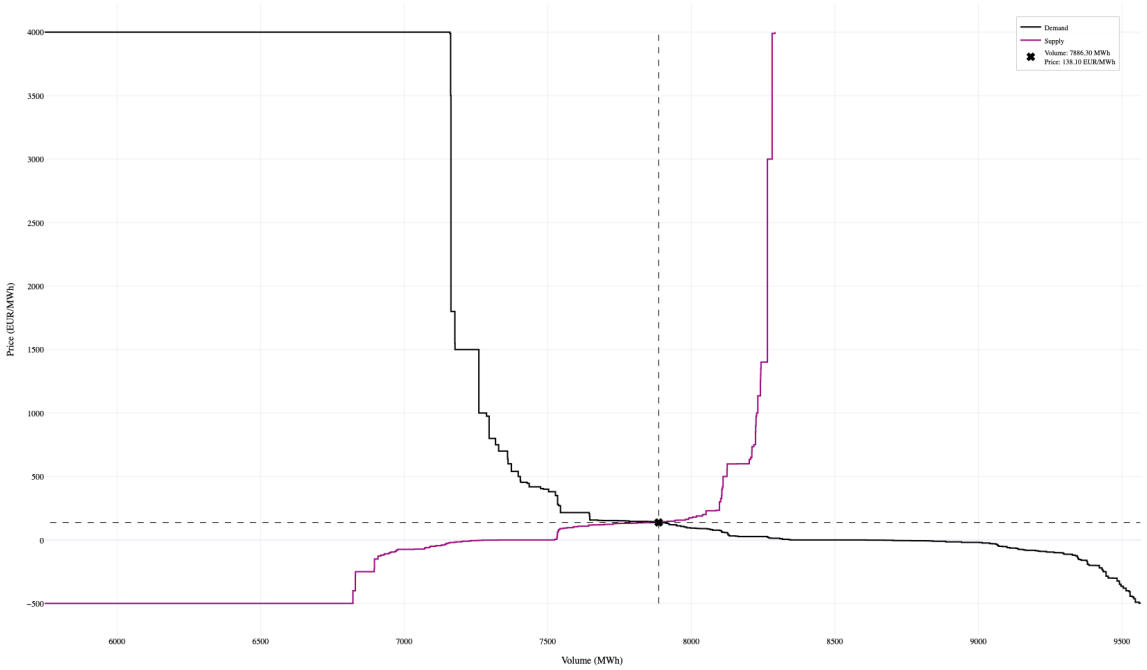


Figure 2.3: Day-Ahead Aggregated Supply and Demand Curves for the Market Area DE-LU, for the Delivery Date: 28th of May, 00:00-00.15.[17]

[17] Based on [Kie19], as of 27.05.2026, 13:19:50 CET/CEST.

2.1.3 The Intraday Market

As the physical delivery of the traded contracts is also mandatory on the intraday market, purely financially motivated participants seek to avoid not-netted positions heavily, as they would have to be balanced out in the balancing market, which is the most expensive one, as it is the last market before delivery. The main purpose of the Intraday Market is to balance out the previously locked in Day-Ahead positions[18] that may deviate due to Power Plant Outages, optimisation of power plant usage (generator), optimisation of demand (costumer), renewable energy producer – changes of forecasts[Kie16]. The CID Market starts on the day before delivery (D-1) at 15:00 CET after closing of the Day-Ahead Market with opening the order books for all 24 hours of the next following day resulting in 24 books in parallel (with the hour 01: 00:00 - 01:00 being the first), 30 Minutes and 15 Minutes. The single resolutions can be traded up until 5 Minutes before delivery. Due to this nature the contract lifetime lasts from 8 hours and 55 Minutes (for the first hour of the next following day D) up until 31 hours and 55 Minutes (for the last hour of the next following day D). Engaged participants therefore can trade up to 32 contracts simultaneously, just considering the hourly resolution.[19] This natural ordering of the hours over the course of the day, also groups them inherently into “peak’ and “off-peak’ hours, stressing their different dynamics during their contract lifetime for trading. Simplified examples for hourly products can be seen in Figure 2.1, a detailed table listing all hourly specificas can be found in the appendix in Table A.1. The main CID market parameters are summarized below in Table 2.1. The pay-as-bid CID market is accompanied by three intraday auctions (IDA), which are single pay-as-cleared auctions, similar to the Day-Ahead market, that take place at specific times each day:

- **IDA 1**: At 15:00 CET on the day before delivery (D-1).
- **IDA 2**: At 22:00 CET on the day before delivery (D-1).
- **IDA 3**: At 10:00 CET on the day of delivery (D).

These IDA serve the purpose of supporting transparency and liquidity pooling, to provide reference prices along the CID contract lifetimes. Similar as the Day-Ahead market, the CID market is also coupled across the different bidding zones, through the so-called Single Intraday Coupling (SIDC), which is a continuous trading mechanism that is based on the XBID architecture, operating on the M7 system of the EPEX SPOT. The IDA auctions are already part of this mechanism, where the XBID architecture collects all orders in a Shared Order Book (SOB). In a window of 20 Minutes before and after, as well as during the IDA gate-closure the continuous XBID market is shut down. Trading on it for a specific contract is finally shut down 30 Minutes before delivery, still leaving trading open for participants of the same delivery-area up until 5 Minutes before delivery. Its official chronological start is therefore at 15:20 CET at the day before delivery D-1. The XBID related market closing times each day:

[18] [See Bal22, p.238]
[19] [See Bal22, p.234]

- **Around IDA 1**: From 14:40 to 15:20 CET on the day before delivery (D-1).
- **Around IDA 2**: From 21:40 to 22:20 CET on the day before delivery (D-1).
- **Around IDA 3**: From 09:40 to 10:20 CET on the day of delivery (D).

that are also dsiplayed in the following figure Figure 2.4: These periods are specifically

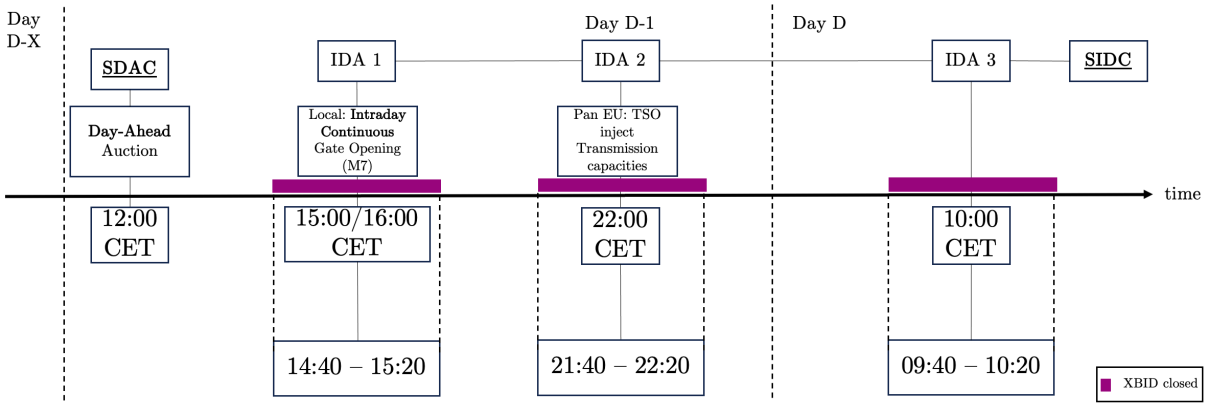


Figure 2.4: Schematic Overview of the IDA Auctions and XBID Closure Times.[20]

important, since this harsh reductions in volumes can be also seen in volume plots later on.

On a european level, the EPEX SPOT and the Nord Pool exchange concentrate the majority of the intraday trading volumes.

Parameter	Value
Tick Size	0.01 EUR/MWh
Minimum Price	-9999.9 EUR/MWh
Maximum Price	9999.9 EUR/MWh
Minimum Order Size	0.1 MWh
Maximum Order Size	1000 MWh
Trading Hours	24/7 (with specific gate closure times)
Trading Days	All calendar days (including weekends and holidays)
Settlement Process	Financial settlement based on the market price at delivery time
Price Grid	Discrete price levels determined by the tick size
Time Grid	Hourly, 30-minute, and 15-minute contracts available for trading
Gate Closure Times	Varying based on the delivery hour, with earlier closures for earlier hours (see Figure 2.1)
Order Types Allowed	Limit Orders, Market Orders, and Cancellations

Table 2.1: Key Parameters of the EPEX SPOT CID Market.[21]

2.1.4 The Balancing Market

As bridge to the previous subchapter: Balancing market exists as to balance out the deviations from the locked in Intraday Market volumes

[20] Based on [Kie19], as of 16.05.2026.
[21] Based on: .

The following table will be provided in the Appendix: Test

Sequence of the market: aggregates progessively better information, participants trade against the arrival of information, microstructure is of each stage is disciplined by the next stages price, sequence also a term structure of price risk

2.2 Microstructure and Organisation of Intraday Power Trading

As the main trading takes place on-exchange and is not broker-facilitated, the core price building mechanism is represented by the LOB.[22] Most academic research is centered around data from highly liquid markets, such as equity or forex markets, where the bid-ask spread is usually only the tick-size as the minimum price-increment, that is possible. Therefore these models often only concern the best bid and aks prices, as there is most volume and a huge market-depth, often not entirely depleted by incoming market orders. In stark contrast, the electricity markets are characterized by a much wider bid-ask spread, that can be several times the tick-size, and a much lower market depth, where incoming market orders often deplete the entire volume at the best bid or ask price, leading to a mid-price movement. This unique characteristic of the electricity markets necessitates a different modeling approach that can effectively capture the dynamics of the LOB and the resulting price movements, which is what this thesis focuses on.

2.2.1 Limit Order Books and Trading Mechanics

The LOB is defined on a fixed discrete grid of prices (price levels), where the step size of each price level is called *tick or tick-size*. This tick-size in the CID market is 0.01 EUR/MWh, which means that the price levels are discretized in increments of 0.01 EUR/MWh. The *tick-size* will be denoted by Δp. The respective minimum and maximum volumes for submitted orders are 0.1 MWh and 1000 MWh, which means that market participants can submit orders for quantities within this range. The order-size will be denoted by V.

The trading hours are 24/7, with specific gate closure times that vary based on the delivery hour, as illustrated in Figure 2.1. All together these parameters are called the resolution parameters of a LOB and are summarised in Table 2.1.

Starting from a basic market setup, participants can submit either Limit Orders (LO) or Market Orders (MO). Participants, sending a MO wish to immediately buy or sell their submitted volume at the best price currently available. Due to this unconditional specification, MOs are also referred to as *aggressive orders*, as they are willing to accept

[22] See [Bun25], where only 0.2 TWh of Intraday Trading was traded on broker platforms vs. 102 TWh on EPEX Spot.

the current market price and reduce the resting volumes at the best bid or ask price. Across the multiple, discrete price levels, the best bid price is the highest price at which there is a buy order, while the best ask price is the lowest price at which there is a sell order.

Definition 2.1 (Best bid and ask prices)**.**

$$\text{Best Bid}(t) = \max\{p_i | q_i > 0, p_i < \text{Best Ask}(t)\} \tag{2.1}$$

$$\text{Best Ask}(t) = \min\{p_i | q_i > 0, p_i > \text{Best Bid}(t)\} \tag{2.2}$$

The natural difference between the best bid and ask prices is called the *bid-ask spread*, which represents the immediate cost of executing a trade aggressively[23] and is a key indicator of market liquidity. The bid-ask spread is defined as the difference between the best ask price and the best bid price, and it reflects the cost that market participants incur when executing a trade. A narrower bid-ask spread indicates a more liquid market, where there is a smaller cost to trading, while a wider bid-ask spread indicates a less liquid market, where there is a larger cost to trading.

Definition 2.2 ((Quoted) Bid-ask spread)**.**

$$\text{Bid-ask spread}_t = P_t^{\text{bid}} - P_t^{\text{ask}} \tag{2.3}$$

From the full bid-ask spread, the half spread can be derived, which represents the cost of trading for a single side of the market, either buying or selling.

Definition 2.3 ((Quoted) Half spread)**.**

$$(H_t)_{t \geqslant 0} = \frac{1}{2} \cdot \text{Bid-ask spread}_t \tag{2.4}$$

This means that incoming MOs can potentially deplete the entire volume at the best bid or ask price, leading to a subsequent mid-price movement. The mid-price is defined as the arithmetic average of the best bid and ask prices and often treated as a proxy for the true underlying price of the asset. It assesses how close the price of a trade is to the ´true underlying market price´.[24]

Definition 2.4 (Mid-price)**.**

$$(S_t)_{t \geqslant 0} = \frac{1}{2} \cdot (P_t^{\text{bid}} + P_t^{\text{ask}}) \tag{2.5}$$

[23] [See CJP15, p.70,] [See Dem68, p.35-36]

[24] [See CJP15, p.71], of course in some situations the true market price may lie closer to either the best bid or ask than the mid-price would suggest.

The best bid and best ask can also be expressed as:[25]

$$\text{Best Bid}(t) = S_t - H_t \tag{2.6}$$

$$\text{Best Ask}(t) = S_t + H_t \tag{2.7}$$

Definition 2.5 (Market Order (MO))**.** A *buy* (respective *sell*) market order is

In contrast, participants sending a LO specify a price at which they are willing to buy or sell a certain quantity of electricity, and these orders are added to the LOB and wait for a matching counterparty to execute the trade. LOs are also referred to as *passive orders*, as they provide liquidity to the market by adding volume to the LOB, rather than immediately executing against existing orders.

Definition 2.6 (Limit Order (LO))**.** A *buy* (respective *sell*) limit order is

How the incoming MO are matched with resting LO to result in a trade, is determined by the matching mechanism of the exchange system. Often this algorithm is based on the price-time priority principle, according to which orders are ranked based on their price and the time they were submitted, just as the name already suggests [CJP15]. Then the MO is matched with the best available LO in the LOB, which is the LO with the highest price for a buy MO and the LO with the lowest price for a sell MO. If there are multiple LO at the same price level, the MO is matched with the LO that was submitted first, following the time priority principle. This process continues until the MO is fully executed or there are no more matching LO available in the LOB. When the volume of the MO exceeds the available volume at the best price level, the remaining volume of the MO is then matched with the next best price level, and this process continues until the entire MO is executed or there are no more matching LO available in the LOB. This process is also referred to as *walking the book*.

If a sell-MO is executed against a buy-LO, it is said to *hit the bid*, while if a buy-MO is executed against a sell-LO, it is said to *lift the ask*.

Besides the discussed basic order types, there are also other modalities available:

- **Iceberg Orders**: These are large orders that are split into smaller visible portions, allowing traders to hide the true size of their order from the market.
- **Stop Orders**: These are orders that become active only when a certain price level is reached, allowing traders to limit their losses or lock in profits.
- **Fill or Kill (FOK) Orders**: These are orders that must be executed immediately in their entirety or not at all, ensuring that traders either get the full quantity they want or no trade occurs.
- **Immediate or Cancel (IOC) Orders**: These are orders that must be executed immediately, but any portion of the order that cannot be filled is canceled, allowing traders to get as much of their order filled as possible without waiting for the entire order to be executed.

[25] [See Luc+21, p.80]

- **All or None (AON) Orders**: These are orders that must be executed in their entirety, but they do not have to be executed immediately, allowing traders to ensure that they get the full quantity they want without having to worry about partial fills.
- **Market Sweep Orders**: These are orders that are designed to quickly execute against multiple price levels in the LOB, allowing traders to quickly fill large orders without having to worry about the time it takes for the order to be executed.

Once an order has been submitted, the trader has the following three actions available, given that the order is placed in the LOB:

- **Cancel**: the order is removed from the LOB and it ceases to exist.
- **Modify**: the trader can modify the price or quantity of the order, and in the case of an iceberg order, the visible quantity. When the price or quantity is modified, the order loses priority in the queue.
- **Deactivate/Activate**: the trader can also choose to 'hibernate' i.e. deactivate the order. In the case of hibernation the order is not visible in the LOB and can not be matched against. The reason why someone would like to hibernate their order could be, for example, in the case of losing a connection to the exchange.

Here the table of the EU Report with Percentages

The information carried by a LOB is a multivariate time series which records the order arrival times and volumes at each price level.[26] In that context, we here explicitly stress its state based representation, for an easier later formulation in modelling approaches. The LOB in every state, can essentially be modelled by the two central inputs, that are the arrival of limit orders and market orders, as these trigger the resting LOs and result in trade executions.[27]

Here the mathematical definition of the LOB:

Definition 2.7 (LOB)**.**

$$\text{LOB}(t) = \{(p_i, q_i, t_i) | i = 1, 2, ..., N(t)\} \tag{2.8}$$

where p_i is the price level of the i-th order, q_i is the quantity of the i-th order, t_i is the time of arrival of the i-th order, and $N(t)$ is the total number of orders in the LOB at time t.

Here the derived optimal depths with a surrounding solid line box:

[26] [See Law15, p.13]
[27] See [Wu+20], p.1

$$
\begin{aligned}
\delta^{\mathrm{SL},*} &= \frac{1}{\kappa^{\mathrm{BM}}} + \big(|q-1| - |q| + 1\big)\epsilon^{\mathrm{BM}} + \big(|q-1| - |q| - 1\big)h \\
&\quad + \alpha(1-2q) - \upsilon\big(t, S^{\mathrm{BM}}, n, h + \epsilon^{\mathrm{BM}}, S^{\mathrm{BM}}\Lambda\big) \\
&\quad + \upsilon\big(t, n, h + \epsilon^{\mathrm{BM}}, S^{\mathrm{BM}}\Lambda\big), \\
\delta^{\mathrm{BL},*} &= \frac{1}{\kappa^{\mathrm{SM}}} + \big(|q+1| - |q| + 1\big)\epsilon^{\mathrm{SM}} + \big(|q+1| - |q| - 1\big)h \\
&\quad + \alpha(1+2q) - \upsilon\big(t, S^{\mathrm{SM}}, n, h + \epsilon^{\mathrm{SM}}, S^{\mathrm{SM}}\Lambda\big) \\
&\quad + \upsilon\big(t, n, h + \epsilon^{\mathrm{SM}}, S^{\mathrm{SM}}\Lambda\big).
\end{aligned}
\tag{2.9}
$$

Also must reference sparse order books as here is the case because other LOB Models are derived from highly liquid underlyings often in quote driven marekts and not order driven markets see here: [BC25]

2.2.2 Order types and classification

As already described earlier, orders can be classified into two main categories, market orders and limit orders, based on their execution characteristics. However, within these two broad classes, the ones of special interest for modelling purposes are those that actually move the mid price. To this *aggressive type*, also cancellations belong, as with a cancellation, the respective order loses its position in the price-time priority queue, leaving the next order in line to become the new best bid or ask price.

Here the categorization into 12 types, where we focus on those types tha actually move the mid price (altering the best bid or ask), see also the descriptive breakdown or orders of a day and their percentage share.

According to [LV19], the main order types in a limit order book (LOB) can be categorized as can be seen in the table below: Explaining the top six *aggressive* order types more closely[29]:

1. The placement of a limit order inside the bid-ask spread. This will cause the price to decrease/in- crease if it is a sell/buy limit order placed below/above the best ask/bid price.
2. The cancellation of the last remaining limit order in the first queue on either the bid/ask side of the book. The cancellation of such a sell/buy limit order will cause the price to increase/decrease.
3. The execution of the last remaining limit order at either the best bid/ask price. This corresponds to the placement of a sell/buy market order at the best bid/ask price that consumes all of the available liquidity at that price and has the same effect on the price as the cancellation of the limit order.

[28] Based on [LV19], as of 16.05.2026.
[29] [See Can20, p.15]

Order Type	Order Arrival Event	Bid Price Effect	Ask Price Effect	Counting Process N
1	aggressive market buy	0	+	$N_1(t)$
2	aggressive market sell	-	0	$N_2(t)$
3	aggressive limit buy	+	0	$N_3(t)$
4	aggressive limit sell	0	-	$N_4(t)$
5	aggressive limit buy cancellation	-	0	$N_5(t)$
6	aggressive limit sell cancellation	0	+	$N_6(t)$
7	non-aggressive market buy	0	0	$N_7(t)$
8	non-aggressive market sell	0	0	$N_8(t)$
9	non-aggressive limit buy	0	0	$N_9(t)$
10	non-aggressive limit sell	0	0	$N_{10}(t)$
11	non-aggressive limit buy cancellation	0	0	$N_{11}(t)$
12	non-aggressive limit sell cancellation	0	0	$N_{12}(t)$

+/0/- indicate whether the order type causes an increase, no change, or decrease in the bid or ask price, respectively.

Table 2.2: Main Order Types in a Limit Order Book.[28]

To describe ONLY the top of the book, the following system can be used (ONLY top book queues) According to [Law15] we can write: $N(t) = (N_1(t), N_2(t), \ldots, N_{12}(t))$ as the multivariate point process of the twelve order types. It further follows that, we can infer the absolute numbers of the shaping variables: $M_a(t)$ as the buy market orders, $M_b(t)$ as the sell market orders, $S_a(t)$ as the ask price and $S_b(t)$ as the bid price. As the tick size, δ, is fixed and each (minimal) price jump is one tick, the relations hold:

$$M_a(t) = N_1(t) + N_5(t) \tag{2.10}$$

$$M_b(t) = N_2(t) + N_6(t) \tag{2.11}$$

$$S_a(t) = S_a(0) + (N_1(t) + N_4(t) - N_{12}(t)) \cdot \delta \tag{2.12}$$

$$S_b(t) = S_b(0) + (N_{11}(t) - N_2(t) - N_3(t)) \cdot \delta \tag{2.13}$$

Therefore the top book spread evolves as:

$$\text{Spread}(t) = S_a(t) - S_b(t) = \text{Spread}(0) + (N_1(t) + N_4(t) - N_{12}(t)) \cdot \delta - (N_{11}(t) - N_2(t) - N_3(t)) \cdot \delta \tag{2.14}$$

In order to avoid a situation where the spread would became less than δ, we impose that the intensities of the aggressive LO additions are conditional on the spread based on the following relationship, where $\mathbb{1}$ is the indicator function:

$$\lambda_i(t) = \lambda_i'(t) \mathbb{1}(S(t) > \delta), \quad i = 3, 4 \tag{2.15}$$

So when the spread is equal to one tick, the intensity of aggressive LO additions drops to zero. Now, it is clear that our model is direction and timing consistent, but since we do not keep track of the depths of the price levels, the model is not volume consistent.

This should serve as bridge on how the different arrivals or counting processes of these order types impact the price. Also at what times the different sites can jump or not. We can formulate the counting processes of all the different event types and how they affect the different system parameters that impact the final mid-price.

2.2.3 LOB Consistency

See here: [Kor25, p.36]: In order for a LOB model to be consistent with the real-world LOB dynamics it must satisfy the following three rules:

1. **Direction consistency**:
    - On the arrival of a marketable buy (sell) order, only the ask (bid) price can move up (down), while the bid (ask) price has to stay unchanged:
    - On the arrival of limit buy (sell) order, only the bid (ask) price can move up (down) while the ask (bid) price has to stay unchanged, given that the LO arrives inside the spread. Otherwise, both the bid and ask prices will remain unchanged:
    - On the cancellation of a limit buy (sell) order, only the bid (ask) price can move down (up) if the canceled order is the only order at the best bid (ask), while the ask (bid) price will remain unchanged. Otherwise, both the bid and ask prices will remain unchanged:
2. **Timing consistency**: the bid and ask prices can only move at times when orders arrive or are cancelled:
3. **Volume consistency**:
    - On the arrival of marketable buy (sell) order whose volume is larger than the available liquidity at current best ask (bid), the price moves up (down) and otherwise stays unchanged:
    - If the volume of the canceled limit buy (sell) order is equal to the volume of volume available at the current best bid (ask), the bid (ask) price moves down (up), and otherwise the price stays unchanged:

One could argue that these rules only apply when no orders can arrive at the same exact moment, but given how the matching engine of the exchange operates the orders are always handled sequentially even in the case that they arrive on the same exact nanosecond. The conditions stated above are necessary in order to model the LOB realistically, regardless of any assumptions about the arrival rates or volume distributions. However, in practice achieving a consistent LOB model is extremely difficult since we would need to keep track of the depths of all the price levels on both sides. In the next section we present the weakly consistent LOB model which only models the best bid and ask queues, and is thus only direction and timing consistent.

2.2.4 Matching Mechanism

2.3 The Market Making Problem

Market Making essentially consists of posting limit orders continuously on both sides of the LOB providing liquidity to the market, at rapid speeds, and earning the Bid-Ask spread as compensation.[30]

The rational behind this way of providing immediacy[31] can be motivated by considering an information based perespective on market participants. A broad classification can be made between *informed* and *uninformed* traders, where the former have access to private information that allows them to make (suspectively) more informed trading decisions.[32] Market Makers (MM) are typically considered to be *uninformed* traders, as they do not have access to private information and instead rely on the public information available in the market. However, they are still exposed to the risk of trading with *informed* traders, who may have an informational advantage and can exploit the MM's quotes to their benefit. This creates a situation of *information asymmetry*, where the MM is at a disadvantage compared to informed traders. These information asymmetries directly translate into the following sources of involved risks a MM faces:

- **Adverse Selection Risk**: This risk arises when the MM trades with informed traders who have access to private information that allows them to make more informed trading decisions. The MM may end up buying at a higher price or selling at a lower price than the true value of the asset, resulting in potential losses.
- **Inventory Risk**: This risk arises from holding an inventory of assets that may fluctuate in value due to market conditions. The MM may be forced to sell at a loss if the market moves against their position, resulting in potential losses.
- **Latency Risk**: This risk arises from delays in receiving and processing market data, which can result in the MM's quotes being stale and not reflecting the current market conditions. This can lead to potential losses if the MM is unable to react quickly enough to changing market conditions.

This perspective is also justified by looking at the different participant types at the EPEX SPOT, where there are not only utilitiy companies but also financial institutions, which are typically considered to be more informed traders, as they have access to more sophisticated trading strategies.

Now balancing these types of risks is the core problem of market making, as the MM must continuously adjust their quotes to manage their inventory and minimize their exposure to adverse selection and latency risks, while still providing liquidity to the market. The

[30] [See Jai+25, p.1,] [See CJP15, p.9 and 20 and also the important specification of MM and Liquidity Privision.]
[31] [See Dem68, p.]
[32] [See CJP15, p.30]

main appoaches on how to come up with optimal strategies for market making can be depicted in the following table:

In this thesis, the considered MM uses LO and MO for maximising expected terminal wealth (cash + inventory) under penalisations for the running inventory as well for terminal inventory, by finding the set of optimal controls $u \in \mathbb{U}$, where $\mathbb{U}$ is the set of admissible controls consisting of a sequence of decision variables.[33]

School of Thought	Core Philosophy & Core Assumption	Key Methodologies & Mathematical Tools	Seminal Frameworks / Examples	Primary Limitation
1) Economics-Based	**Perfect Rationality**: Agents maximize utility under structural constraints. Focuses on *why* liquidity is priced based on risk aversion and asymmetric information.	• Dynamic programming • Game theory (Nash / Stackelberg) • Expected utility functions	• **Inventory Models**: Ho-Stoll, Avellaneda-Stoikov • **Information Models**: Glosten-Milgrom, Kyle's lambda	Highly stylized; yields closed-form solutions that fail to capture real-world, high-frequency order book dynamics.
2) Physics-Based	**Zero Intelligence (ZI)**: Complex macro market allocations emerge from the statistical mechanics of non-strategic, random order flows.	• Stochastic processes • Poisson arrival rates • Master equations / Particle physics analogies	• **ZI-LOB Models**: Anufriev-Panchenko, Maslov model • **Statistical Mechanics**: Santa Fe Institute models (Farmer et al.)	Completely ignores strategic behavior; agents do not adapt to adverse selection or toxic order flow.
3) Mathematically-Based	**State Dependence**: Focuses strictly on data-driven, conditional state transitions of the actual limit order book queue.	• Markov chains / Jump processes • Point processes (Hawkes processes) • Queueing theory	• **Queue-Reactive Model** (Huang, Lehalle, Rosenbaum) • **Cont-Stoikov-Talreja** Markovian framework	High dimensionality; suffers from the "curse of dimensionality" when scaling up the number of tracked book levels.

Table 2.3: Frameworks of Market Making Microstructure[34]

For a more detailed overview of the main schools of thought and connected concrete concepts, see [Jai+24b].

The resulting concepts and models based off of these theoretical strands often lack real world applicability, as they are based on highly stylized assumptions or just miss important real world caveats.

Here stole everything from: [Law15] page 5: In this section, we highlight some issues of existing market-making models in the con- text of high-frequency trading. 1. In modern

[33] [See Luc+21, p.]
[34] Based on: Own Concept.

financial exchanges, prices are only allowed on a predefined fixed grid called price ticks. As a result, price is a pure-jump process and it has two dimensions, 5 namely times and magnitudes of the jumps. Diffusion can only approximate the magnitudes of the jumps but cannot describe the properties related to timing of the jumps such as jump clustering. 2. The common assumption of Poisson order arrivals is often rejected in empirical lit- erature as order arrivals depict strong self-excitation behavior [18–20]. 3. All models assume that price and order arrivals are independent, which is far from the truth by realizing that price rises with large buy market order and falls with large sell market order. Because of adverse selection [1, 2], the absence of this crucial dependency structure will generate large phantom profit for the market maker and cause the average profit of the market-making strategy to be overstated. 4. Since almost all exchanges nowadays use the price-time2 priority, changing price or quantity of limit orders means loss of priority. Nonetheless, existing models all use regular control to continuously adjust the quotes without any penalty. 5. A critical component of many existing models is the demand/supply rate function. For example, in [8], it is in the form while in [9], it can be expressed as . Yet the parameters are hard to estimate since when the limit order is more than one tick from the best quote, the execution probability is minuscule (e.g. less than 36. For the sake of simplicity, existing models assume all orders are of the same size. Such an assumption will mask the risk of overtrading of the market maker. For instance, to continuously maintain priority in the queue, the market maker may post more limit orders in the order book than his risk tolerance. However, the arrival of one giant market order may raise his inventory to an unacceptable level, which can potentially lead to bankruptcy. Such kind of risk cannot be modeled with all orders having the same size. In contrast to other

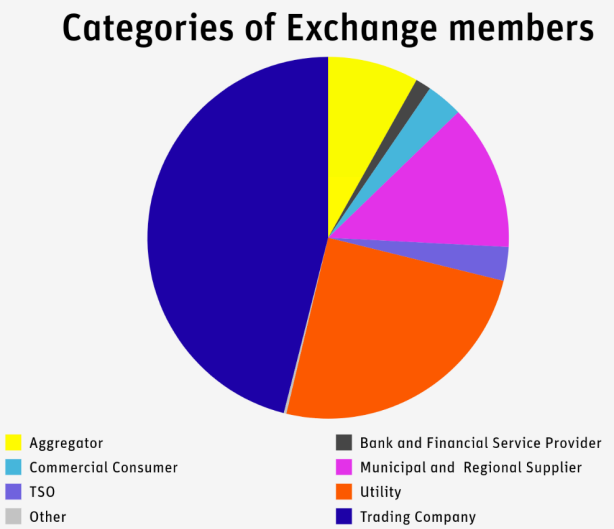


Figure 2.5: EPEX SPOT exchange member categories.[35]

financial markets, the trading on the EPEX SPOT does not involve designated market

[35] Based on [Kie19], as of 31.07.2026, 17:11:50 CET/CEST.

maker (DMM) programs, where specific participants are incentivized to provide liquidity to the market.[36]

In general, the main theoretical approaches to tackle this problem can be categorised into the following streams:

1. Physics based zero intelligence models
2. Economics based rational agent models
3. Math based stochastic optimal control models

A good overview with a more granular perspective can be found in [Jai+24b] where great emphasis is placed on the modelling approaches.

The classical problem formulation based on Ho & Stoll (1981) is to choose the optimal depth size for posting LO into the book (See [See CJP15, p.29])

Also mention the advantage of potentially getting a fee rebate from the exchange in contrast to the taker fee that liquidity takers have to pay for their orders

Key problem: Balancing the optimal limit order posting depth and the fill probabilities Latency risk as the EPEX Spot is not offering co location services, so the MM is exposed to latency risk, as they are not able to react as fast as other market participants that are co-located with the exchange servers. This can lead to a situation where the MM's quotes are stale and do not reflect the current market conditions, resulting in potential losses.

Use stochastic optimal control to tackle the classical problem of solving for the optimal depth as distance from the midprice for posting limit orders, also core problem: how to get the fill probabilities or approximate them

[36] [See Bal22, p.230]

3 Empirical Microstructure: LOB Characteristics and Market Quality

In the following chapter, so called *quality-parameters* of the LOB will be analyzed, which are the key empirical characteristics of the LOB. These will be used to first motivate, then calibrate and finally evaluate the mathematical model for the derivation of the optimal market making strategy. These *quality-parameters* include the following aspects, which we defined mathematically earlier in section 2.2 Microstructure and Organisation of Intraday Power Trading:[37]:

- Interarrival times
- Volumes and Volatility
- Spread measures [38] and the Half Spread
- Market Depths and Trade sizes
- Market Impact
- Message activity and the relationship between cancellations, executions and distance from the midprice

As these components are important drivers of overall *liquidity*, they are key in forming the shape of the LOB, what will be important for later modelling considerations in chapter Mathematical Modelling. Before we start to analyse the empirical characteristics of the LOB, we will first provide a brief introduction of the used data.

3.1 Data and Data Processing

In order to collect all L3 market-by-order events for the delivery area germany, we collect all single order books from the EPEX SPOT for the TSOs: 50Hz, Amprion, Tennet and TransnetBW for the time range of the full quarter of Q2 2026. The single observations are timestamped on millisecond level, which is the highest possible granularity that can be achieved with the EPEX SPOT L3 data. A more detailed data description can be found in the Data Appendix under LOB Data Description. The data contains all limit order additions and cancellations as well as market orders for each one-hour product. Due to the XBID mechanism, the data also includes orders from other delivery zones, but the origin of the orders is not included in the data. In the following table, a first descriptive breakdown with emphasis on the average events per day is provided, that serves as first basis for the more detailed analysis of the intraday activity.

[37] [See CJP15, p.61.]
[38] [See Bal22, p.229]

Hour	Year	Number of (Trading) Days	Average Events per Day
0	2019	365	X
1	2019	365	X
...	...	...	...
23	2019	365	X

Table 3.1: Data Overview: Hourly Event Counts and Averages.

3.2 Intraday activity patterns

Here just follow the Cartea Book for all the setcions, investigate data and also cite the literature for these findings here for confirmation, Lead Thesis did it as well like that

3.2.1 Interarrival Times

As already mentioned in XX, when applying a more granular analysis, the modelling of prices requires a *discrete*, rather than *continuous* viewpoint.[39]

In the below figure we can see, that the interarrival times seem to be quite good approximated by a power law distribution with very heavy tails.[40] In the below figure, we can see

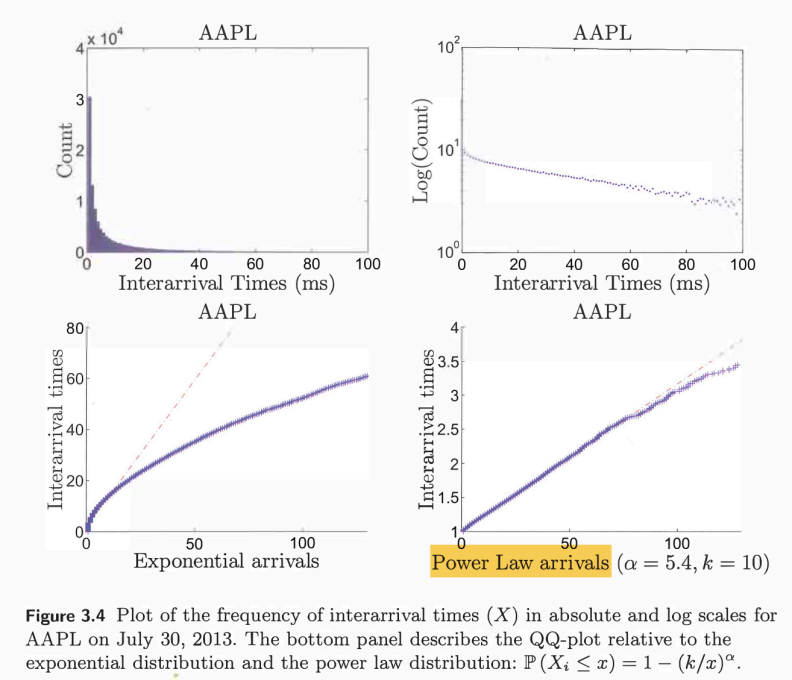


Figure 3.4 Plot of the frequency of interarrival times (X) in absolute and log scales for AAPL on July 30, 2013. The bottom panel describes the QQ-plot relative to the exponential distribution and the power law distribution: $\mathbb{P}(X_i \leq x) = 1 - (k/x)^{\alpha}$.

Figure 3.1: Interarrival Times.[41]

[39] By interarrival times, we refer to the interval times between movements at either the bid or the ask, [see CJP15, p.48].
[40] [See CJP15, p.48]

that changes to the bid- and ask are not independent, but have a strong autocorrelated component, suggesting that they appear in *clusters*.[42] Rapid changes in the bid/ask are followed by further rapid changes, while a relatively long calm period is also similarly followed by another calm period.[43] From this we can conclude that the interarrival times are

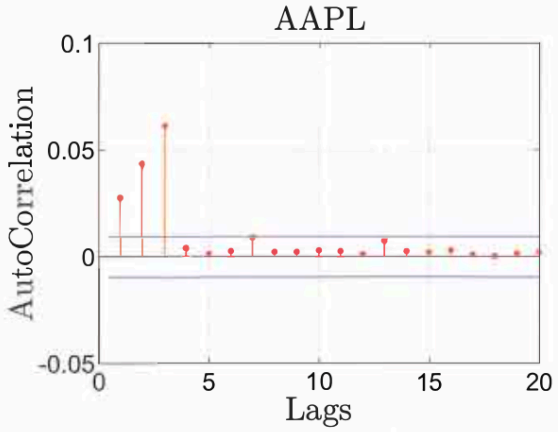


Figure 3.5 Plot of the autocorrelation function for interarrival times (AAPL, July 30, 2013).

Figure 3.2: Interarrival Times ACF.[44]

not independent and identically distributed (i.i.d.), but rather exhibit a strong temporal dependence structure. This has important implications for modeling the dynamics of the limit order book, as it suggests that past events can provide valuable information about future market activity.

3.2.2 Volumes and Volatilty

The volume pattern over a trading session or lifetime of a contract for the intraday power markets is quite distinctive from other financial markets, that typically show a U-shape over the course of a normal trading day. Starting off with high volumes near market opening, a slight volume lull in the middle of the day, and a peak in volume near market close.[45] This shape generally reflects the quote updates based on newly arrived information. In contrast, the intraday power markets show a different pattern, with a peak in volume near gate closure, but also a significant increase in volume during the last hour of trading.[46]This is due to the fact that market participants have more information about the supply and demand conditions for the delivery period as it approaches, and they adjust their positions accordingly (*trading against information arrival*). It can also be seen clearly

[41] Based on [See CJP15, p.49].
[42] [See CJP15, p.49]
[43] [See CJP15, p.49]
[44] Based on [See CJP15, p.49].
[45] By volume we refer to the trading volume measured in MWh.
[46] With activity, also the volume keeps up, as 80% of total volume is traded within the last three hours of a trading period [See Bal22, p.238].

that the volume picks up at around 6 to 7 hours prior to gate closure.[47]. Also algorithmns may try to close out any remaining residual volumes at the end of the trading sessions, what possibly leads to an increase in volumes towards gate closure in combination with execution strategies that try to postpone non-urgent transactions to times when spreads are narrow and more preferable, what is also the case near gate closure.[48] This would match the observed tightening of the spread that can be seen in 3.2.3 Spreads. Other distinctive

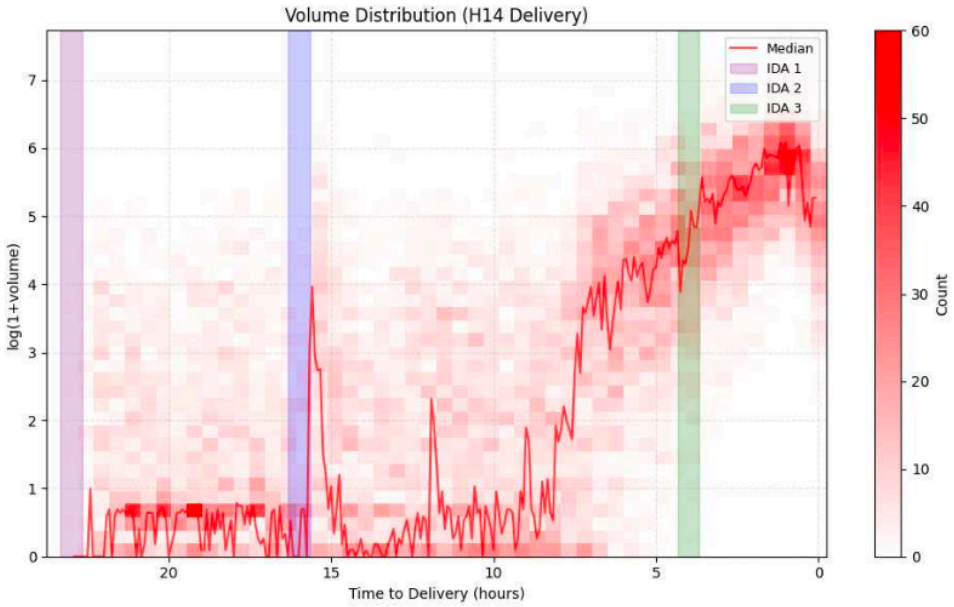


Figure 3.1: Volume distribution for hour 14 delivery.

Figure 3.3: Volume distribution for hour 14 delivery.[49]

features of the volume pattern are the three spikes in volume that occur shortly after the IDAs. During the IDAs and the final hour of trading, XBID is closed, meaning that the LOB only contains orders with delivery at one of the four German delivery zones.[50] The effect of this can be seen in the figure as the executed volume drops visibly during the IDAs and spikes right after when the LOBs are joined together and overlapping orders are executed. The Volume also seems to exhibit a susbtantial variability, as can be seen in the below figure. To focus on the time-to-delivery dependency of the traded volumes, we see in the below functional approach heat maps (left panels) of daily volume for the year 2013 for AAPL and MENT. The heat-maps are generated by first bucketing traded volume into five-minute windows throughout the day, for every day of the year. Then,

[47] This is usually the time when the morning shift starts for traders in Europe, as the market is open 24/7, the trading is done in three shifts: morning, evening and night shifts or alternatively, the night shift might be working from a country with a more suitable time-zone like Singapore
[48] [See CJP15, p.66]
[49] Based on [See CJP15, p.63].
[50] The schematic timeline can be followed in Figure 2.4.
[51] Based on [See CJP15, p.63].

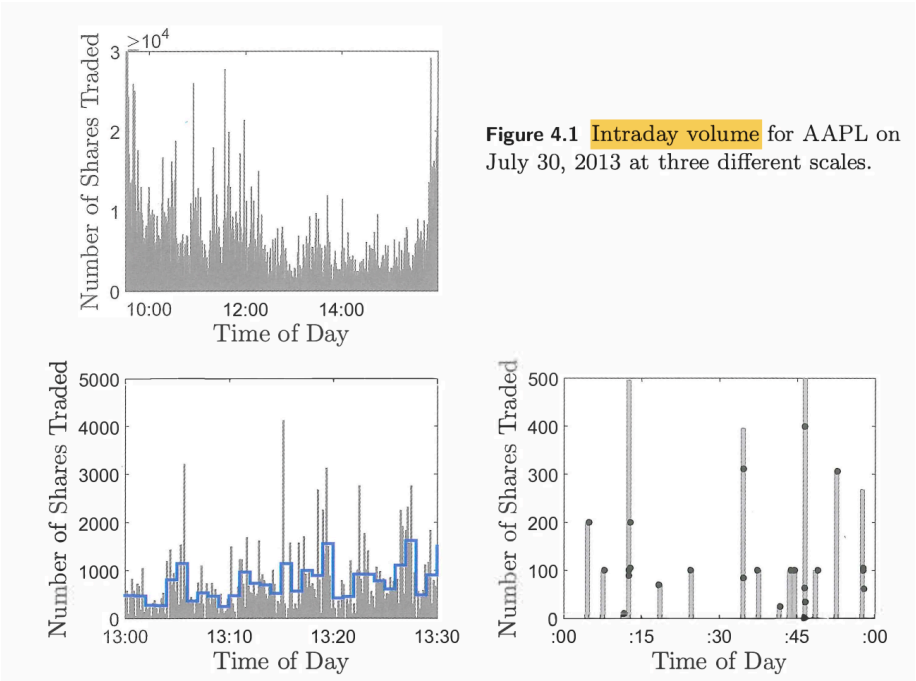


Figure 4.1 Intraday volume for AAPL on July 30, 2013 at three different scales.

Figure 3.4: Intraday Volume.[51]

for each five-minute bucket, we compute the distribution of volume. The heat-map is a visualisation of the collection of these distributions for each five-minute bucket all at once. In the figures we use coloured lines to represent the first and third quartiles, as well as the median.[52] A clear dependency of the volume on the time-to-delivery can be seen, as descibed earlier but here with the advantage of being able to spot outliers (right panels). Volume itself seems to have significant time-series persistance and positive and significant 'correlation' with volatility.[54] Intraday volume has very large peaks.[55] This is evident by further following the time to delivery timescale but graphing the volatility in the below figure.

In the above figure we can see the large accumulation of transactions whose prices end in exact dollar-values prices, with the most pronounced spike at the round 50 cent value, with slightly less pronounced spikes at the round 10th of cents values.[58] As incoming aggressive MO just hit standing LO in the LOB, is is ultimately the MMs who deciide to post their orders at these round values for providing liquidity.[59]

Further going into more details and dividing the tradable hourly products into peak

[52] [See CJP15, p.65.]
[53] Based on [See CJP15, p.65].
[54] By volatility we refer to the standard deviation of the price changes measured in €/MWh.
[55] [See CJP15, p.64]
[56] Based on [See CJP15, p.63].
[57] Based on [See CJP15, p.69].
[58] [See CJP15, p.64]
[59] [See CJP15, p.64]

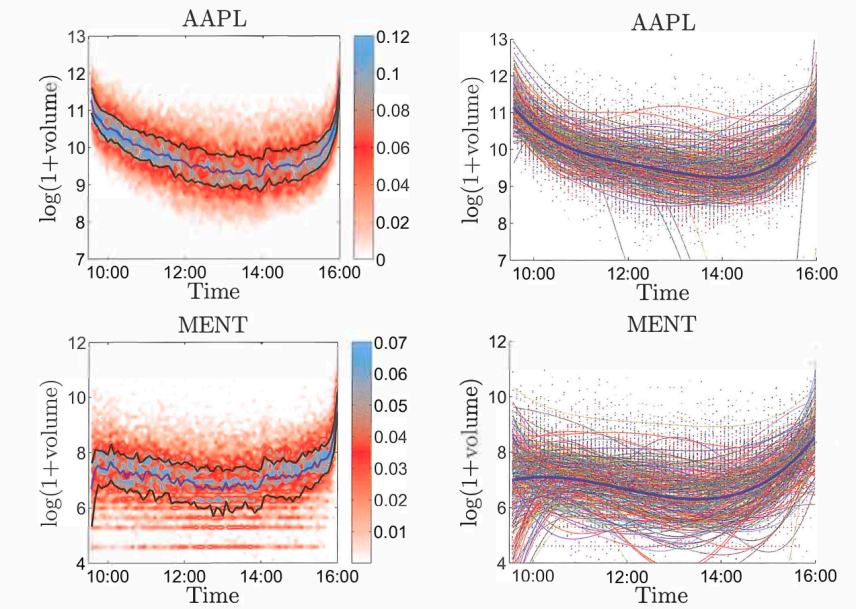


Figure 4.2 Volume as a function of the time of day.

Figure 3.5: Intraday Volume as function of the time of day.[53]

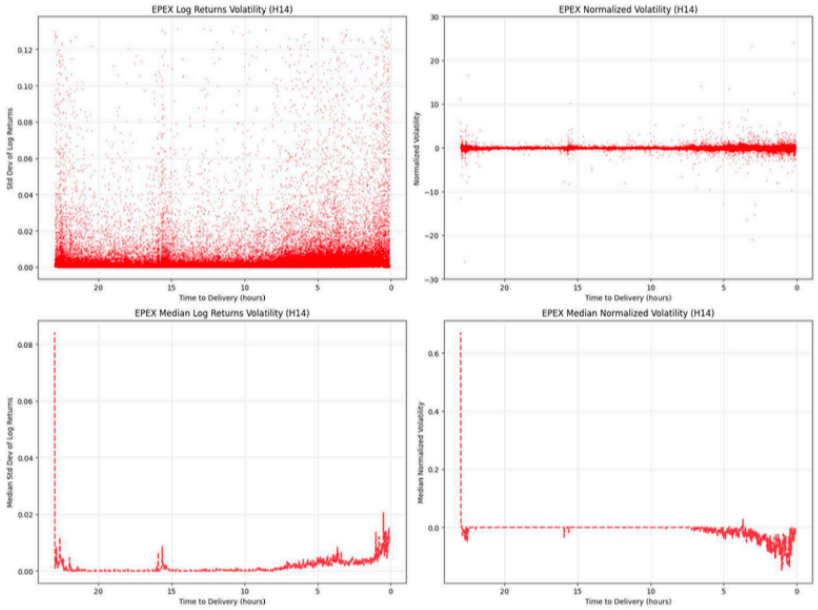


Figure 3.3: Volatility distribution as a function of time to delivery.

Figure 3.6: Intraday Volatility.[56]

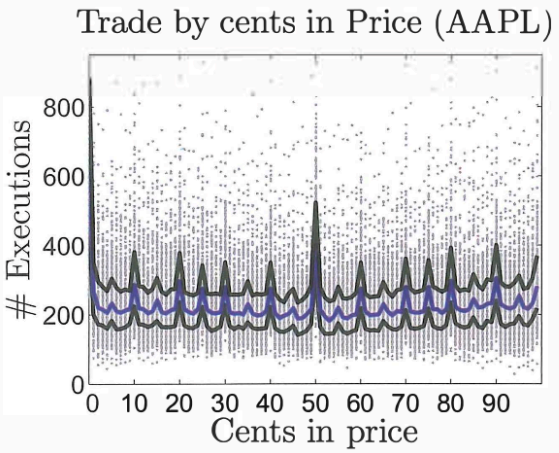


Figure 4.5 Price pattern: frequency of executions by number of cents in the price.

Figure 3.7: Price pattern with cent-level bounds.[57]

and off-peak products and also analysing potential weekend vs. during the week effects, we can see that the volume pattern varies for these subtypes. Samuleson Effect here From previous intraday pattern: Baseline intensity has to be dependent on the remaining contract lifetime, therefore being non-stationary

Percentage table for the different order types to stress that onyl few types are relevant and also focus on only the best 3 price levels because mostly only 1 tick moved (?).

3.2.3 Spreads

Spreads measure the immediate cost of executing a trade aggressively.[60] In comparison to the later analysed *market impact*, that centers more around larger trades, spreads measure the execution costs of smaller trades in relation to the current 'market price'.[61] As can be seen in the figure above, the bid-ask spread over a trading session seems to follow an L-shape pattern, in line with the findings of: XX. The spread decreases with time as there is more activity closer to the delivery. This is consistent with the findings of [See Bal22, p.238]. During the IDAs the spread increases significantly, which is expected given that the spread is negatively correlated to volume according to XX. The median spread of the average trading session is 0.89 €/MWh, which is considerably less than what was found in previous studies. For comparison, the median spread in 2015 reported by Balardy was about 3.52 €/MWh. This decrease can be attributed to the annually increasing volume in the intraday market. Although the median spread is more than three times smaller than the median spread in 2015, it is still 89 times the tick size, which compared to equity markets is massive. In equity markets the spread is typically few times the tick size at most. The weekend effect in liquidity can also be observed through the spread as Balardy reports

[60] [See CJP15, p.70]
[61] [See CJP15, p.71]
[62] Based on [See CJP15, p.51].



Figure 3.2: Bid-ask spread distribution as a function of time to delivery.

Figure 3.8: Bid-ask spread distribution as a function of time to delivery.[62]

in that the spread is on average 13% higher during the weekends. The tick size also has a large effect on the spread distribution. Given that the tick size of 0.01 €/MWh is quite small compared to the average price per megawatt, the observed distribution is not that surprising. Not surpising, the spread is also connected to volumes, as the more frequent the trades, the better the spread, and also to the volatility of the spread itself. More frequently traded assets tend to have less volatile quoted spreads [Bal22]. A slightly older study conducted by [HW13] found a lower mean bid-ask spread of about 2.97€/MWh. As the durations of the trading sessions per product differ in their lengths, a smooting of the bid-ask spread can be observed, the longer the trading session lasts [See Bal22, p.238]. Concluding, the bid-ask spread generally follows more like an L-shape instead of a U-shape pattern as is usually observed in other financial markets [See Bal22, p.238]. As for the main drivers of the bid-ask spread, increased volatility positively affects the bid-ask spread, especially pronounced for off-peak produtcs [See Bal22, p.246]. Not surprisingly, fundamental variables like load, wind, solar and their associated forecast errors (positive and negative) have a negative impact for the bid-ask spread [See Bal22, p.247]. The Bid-ask spread has two important dependency relationships: First, the impact of the Bid-ask spread on posted LO depths can't be ignored, since the bid-ask spread is rarely tick size and often exceeds it by several magnitudes.[63] Second, it is relevant for the arrivals of MO, as we have seen that they appear in situations where the spread is favorable for them and narrow. Pictured in the figure above, quoted spreads are initially high, decline rapidly during the first half-hour of trading, and are mostly constant throughout the remainder

[63] [See Luc+21, p.78]
[64] Based on [See CJP15, p.75].

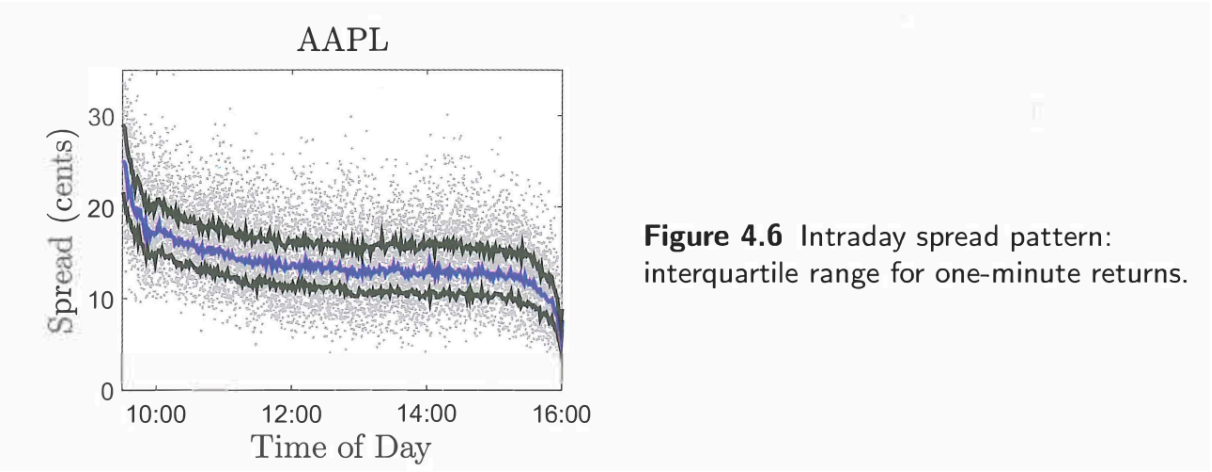


Figure 3.9: Intraday spread pattern (IQR).[64]

of the day until the last (half) hour of trading, when the spread rapidly declines. This pattern in quoted spreads is also seen in the effective spread.[65] We can resonate that during a period with a constant flow of information more trading and lower spreads occur together.[66]

3.2.4 Market Depths and Trade Size

Confirmed here as already presented in 2.3 The Market Making Problem, market depths and associated trade sizes are not independent of another.[67] Depth in this context refers to the available volumes resting on the price levels in the LOB at a given time, ready for immediate execution.[68] This behaviour can be motivated as market participants look to minimize their market impact, splitting larger block trades into several smaller ones. This indicates two phenomens: First, as already stated, the depth of the LOB and size of the incoming orders are highly correlated. But also secondly, this splitting induces, and at least in parts, explains the already observed clustering of MO arrivals. To get an intuition for the resting depths of the LOB, the figure below shows that the available liquidity on both sides is relatively similar and centered around 5 megawatt quantities. Five megawatts is also the minimum visible quantity for an iceberg order on EPEX SPOT, which might explain the high number of orders with such a quantity. Interestingly, the volume on both sides drops visibly after the third IDA. This is most likely caused by execution algorithms getting more aggressive towards the end of the trading session as they split orders into small pieces and try to execute passively in order to get a better price. Verifying this is of course not possible since all order information is anonymous. Looking beyond just the best bid and ask levels, the figure below shows the depths for the

[65] [See CJP15, p.75]
[66] [See CJP15, p.75]
[67] [See CJP15, p.80]
[68] [See CJP15, p.79]

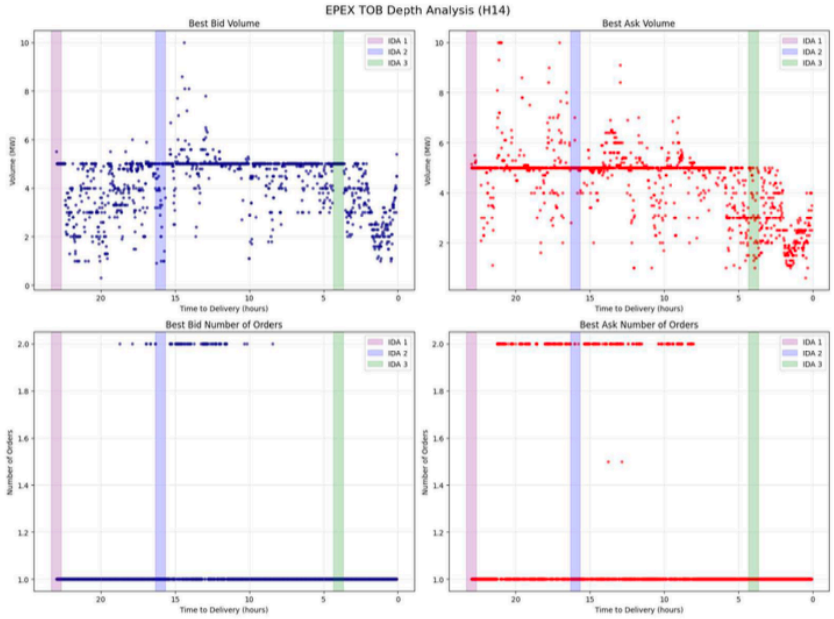


Figure 3.4: Top of the book depth as a function of time to delivery.

Figure 3.10: Top of the book depth as a function of time to delivery.[69]

best five price levels on both sides. In combination with the previous figure, it is evident, that the number of individual orders at each price level is almost always one or two, which means that the order queues are very weak. For market making purposes strong queues would be optimal since then the price impact and the resulting negative effect on the market makers inventory would be smaller. Figure 3.5, which shows the cumulative liquidity and the total number of the orders at the top five levels of both sides of the LOB, confirms the earlier observation that towards the end of the trading session the orders get smaller and the liquidity at the top levels of the book gets more scarce. In the figure below, we plot the mean and median distance between the top price levels over the average trading session. The distribution for both sides is very similar to the bid-ask distribution, excluding the IDAs. The distances get smaller towards delivery which would indicate that the total liquidity in the LOB increases. During the IDAs and the final hour of trading, the distances jump up as a natural result of the XBID being turned off. As the arrivals of MO depend on the Bid-ask spread in favor for tighter conditions, we can also conclude that the posted LO depths are dependent as well. Market depths correlate negatively with higher bid-ask spreads [see Bal22, p.237], what is especially pronounced for off-peak hours and weekeends. Higher bid-ask spreads primarily result from higher sell prices and lower buy prices [Bal22, p.237], leaving business days and peak-contracts as the most liquid combinations in the market. As the spreads widen, the MM tend to post their LOs closer to the midprice, which is also a natural reaction to the lower liquidity in

[69] Based on [See CJP15, p.75].
[70] Based on [See CJP15, p.75].
[71] Based on [See CJP15, p.75].

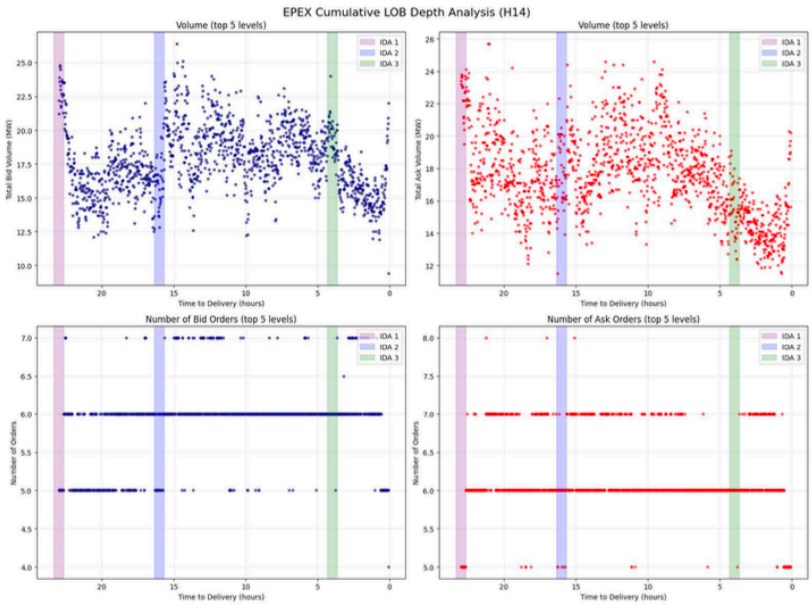


Figure 3.5: LOB depth for top 5 price levels as a function of time to delivery.

Figure 3.11: LOB depth for top 5 price levels as a function of time to delivery.[70]

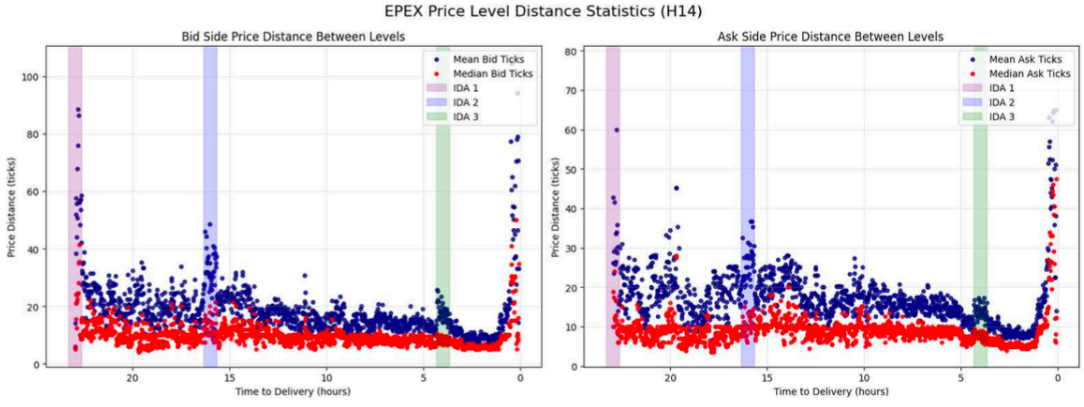


Figure 3.6: Mean and median distance between the top 5 price levels as a function of time to delivery.

Figure 3.12: Mean and median distance between the top 5 price levels as a function of time to delivery.[71]

the LOB.

3.2.5 Market Impact

As we have seen from the previous market-depth analysis, for the CID market, the market impact plays a big role since resting LO volumes at the best bid and ask are rather small and would be entirely consumed quickly by large incoming MO, depleting these price levels, 'walking the book' and attaining a stepwise worsening average price. Price impact measures the cost of executing larger trades via their impact of trading on prices.[72] A linear relationship between price changes and net order flow is plausible for a wide range of midprice changes, what justifies the use of a simple linear model to measure the market impact. By the simple linear model, we can see that there is a positive relationship between net order flow and price changes. That is when net order flow is positive (negative), that is more (less) buy than sell MO, the midprice tends to increase (decrease).[73] The price impact can be further decomposed into two components: the permanent and the temporary impact. An exemplary figure of the results for the temporary price impact can be seen below: The third picture in the Figure shows how the slope of this linear impact

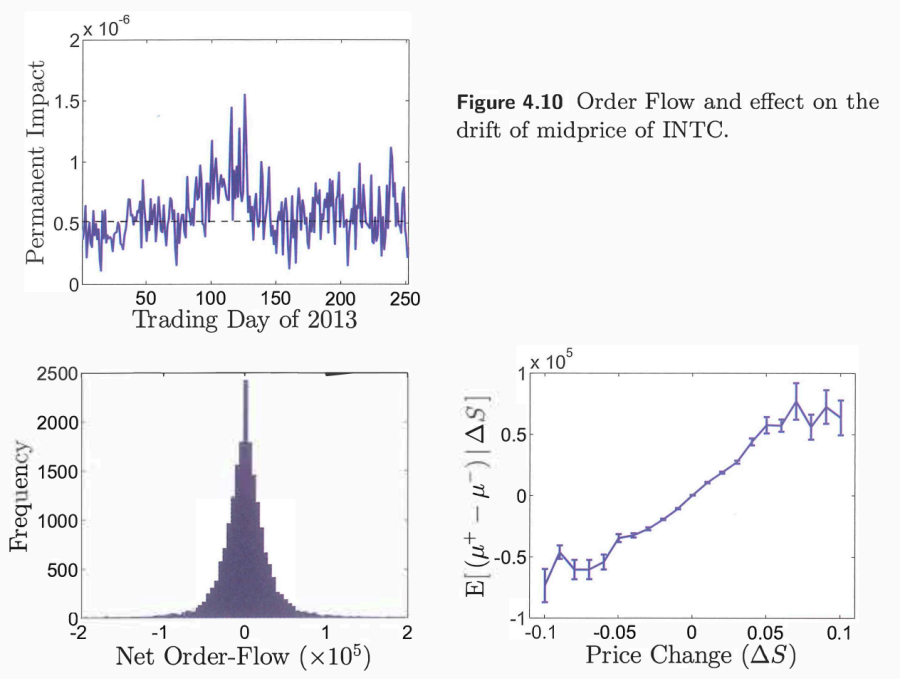


Figure 4.10 Order Flow and effect on the drift of midprice of INTC.

Figure 3.13: Order Flow and effect on the drift of midprice of INTC.[74]

model fluctuates throughout the entire day. We see that the largest impact tends to occur in the morning, then this impact flattens and stays flat throughout the day, and towards

[72] [See CJP15, p.88]
[73] [See CJP15, p.88]
[74] Based on [See CJP15, p.75].

the end of the day it lessens. Such a pattern is seen in a number of assets and is consistent with the reduction in spreads and increases in depth we have documented earlier. So, our very preliminary and limited analysis suggests that order arrival seems to be followed by further order arrival on both sides of the book, and more on its own side than on the other side of the book. As we will now see, MOs do not arrive at random times. They tend to arrive when spreads are narrow, and opportunistically hit orders that are posted closer to the mid price, so it is only natural that we should observe a wider spread after an execution.[75]

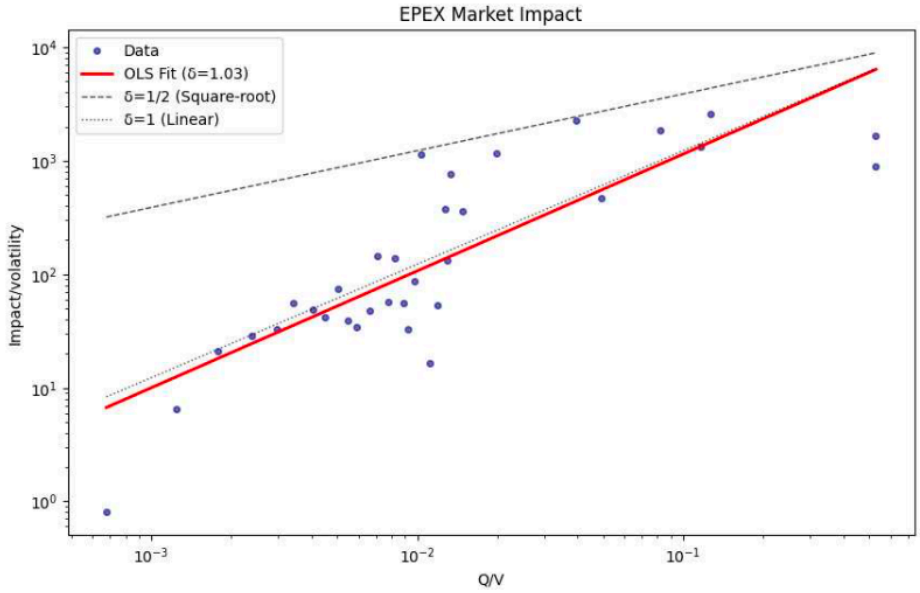


Figure 3.10: Approximation of the market impact of single MOs.

Figure 3.14: Market impact.[76]

3.2.6 Messages and Cancellations

Key components forming the shape of the LOB are the arrival and cancellation rates of LOs and MOs. As these are critical inputs for the overall optimisation problem of the MM, in the sense that their intensities determine the frequency of the LOs being lifted and the resulting gain from the spread. Descriptive statistics are provided in the table below: Analysing the arrival rates of MOs and LOs, Figure 3.7 shows the average intensity of MO arrivals during a trading session for hour 14 delivery. The distinction between aggressive and non-aggressive MOs is that aggressive MOs move the current mid-price i.e. the volume

[75] [See CJP15, p.87]
[76] Based on [See CJP15, p.88].
[77] Based on [See CJP15, p.91].
[78] Based on [See CJP15, p.91].

Asset	Mean	StdDev	P01	Q1	Median	Q3	P99
ISNS	1,711	6,078	173	450	760	1,745	8,943
FARO	24,038	10,871	8,524	16,277	22,347	29,232	71,445
MENT	59,661	21,755	23,157	43,477	53,972	72,639	131,715
AAPL	531,728	166,652	280,242	417,576	500,680	614,437	1,067,248

Table 4.14 Daily Number of Messages (in 000s).

Figure 3.15: Number of messages per day.[77]

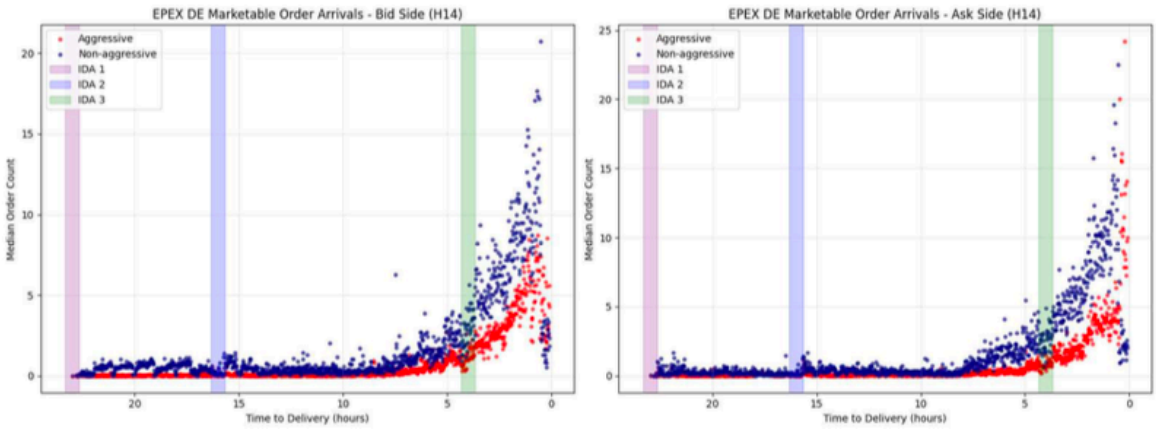


Figure 3.7: MO arrival distribution as a function of time to delivery.

Figure 3.16: MO arrival distribution as a function of time to delivery.[78]

of the MO is large enough to deplete all the liquidity at the current best bid or ask, and non-aggressive MOs have less volume than is currently available at the best bid or ask. The intensity of MO arrivals is highly correlated with the volume pattern in Figure 3.1, but the pattern is smoother and follows an exponential shape. The intensity drops significantly only in the final hour of trading, and the IDAs seem to have little to no impact on the intensity. The share of aggressive MOs to non-aggressive MOs seems to be constant. Von luckner and Kiesel also provide strong empirical evidence about the clustering of MOs in the German intraday market in. Clustering means that successive orders are more likely to be of the same sign, which can be attributed to market participants splitting their MOs into smaller pieces, following the behavior of other traders or acting on the same information in timely proximity. The ability to cancel and place new limit orders is critical for the market maker in order to change their LO placement in the LOB upon arrival of new market events. Figures 3.8 and 3.9 present the intensities of LO arrivals and cancellations for the hour 14 delivery. Aggressive LO arrival is defined as a LO that arrives inside the spread thus changing the mid-price and non-aggressive arrival is an order that arrives at the current best bid or ask, or a worse price. Aggressive cancellation is defined as a canceled order that moves the mid-price which happens when the canceled order is the only order at the best bid or ask. The patterns for LO additions and cancellations are almost identical, which is not a coincidence: most of the orders are placed and then canceled in the next few seconds. The patterns for additions and cancels are not as smooth as the MO arrival pattern and are significantly affected by the IDAs. During the IDAs, the intensities for both additions and cancellations drop significantly and jump right after when XBID is turned on again. The drop in intensity in the final hour is also very clear. The share of aggressive LO additions and cancels is relatively constant with small spikes during the IDAs due to lower liquidity in the LOB.

From the above table we find that the distance to the midprice is (naturally) shorter, and we can compute that 36% of orders were executed at prices between 1 and 3 cents from the midprice, and 42% at prices between 4 and 6, so that only 22% of executions were relatively far from the midprice (7+ cents away).[81] In the above figure of total executions, as the distance from the price at which the original LO was posted increases. This represents an approximation to the 'fill probability' - the probability that a posted order is executed, what we will also later use in the modelling. Assuming a certain market share of ρ that the MM of the total traded volume has, or wishes to achieve, and that is chosen in advance, we can use this directly in the sense of[83] multiplying the effective arrival rate of market orders hitting his limit orders by ρ. For example, when ρ=5%, that means the effective arrival rate of market orders hitting his limit orders is around 5% of all market orders. He may achieve this by continuously adjusting his limit order quantity to be 5% of the total quantity in the queue but the detailed mechanism is outside the

[79] Based on [See CJP15, p.91].
[80] Based on [See CJP15, p.93].
[81] [See CJP15, p.93]
[82] Based on [See CJP15, p.94].
[83] [See Law15, p.9]

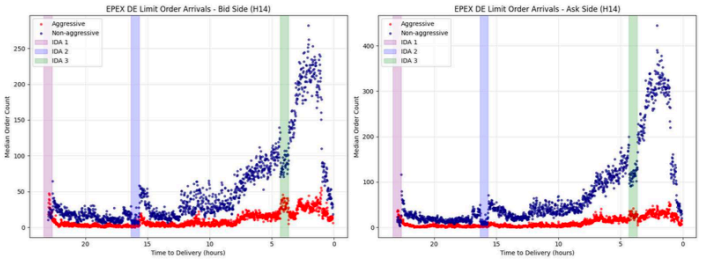


Figure 3.8: LO arrival distribution as a function of time to delivery.

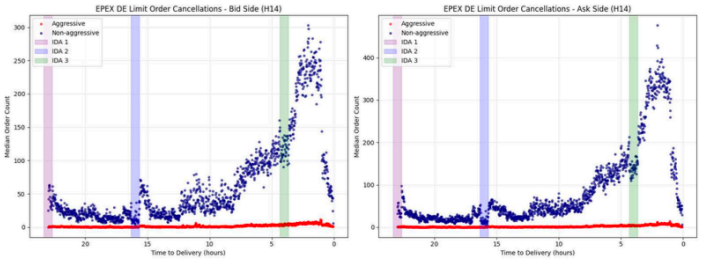


Figure 3.9: LO cancel distribution as a function of time to delivery.

Figure 3.17: LO and CA arrival distribution as a function of time to delivery.[79]

Distance to Midprice	At Post			At Exit		
	Posts	% Exec	Exec	Posts	% Exec	Exec
< 2	905	78.6	3.3	988	88.4	4.1
2	3,053	64.7	9.2	3,508	76.4	12.5
3	5,193	55.4	13.5	6,236	67.5	19.7
4	5,617	44.4	11.7	6,448	51.5	15.6
5	6,374	34.9	10.4	7,557	45.5	16.1
6	7,626	27.6	9.8	7,586	29.9	10.6
7	7,996	20.2	7.6	7,624	20.4	7.3
8	7,826	15.9	5.8	8,062	14.2	5.4
9	7,675	12.3	4.4	7,946	7.5	2.8
10	7,967	8.6	3.2	7,487	6.1	2.1
> 10	195,415	2.3	21	192,205	0.4	3.8

Table 4.17 Messages by distance to midprice at post and at exit (AAPL 2013-07-30).

Figure 3.18: Messages by distance to midprice.[80]

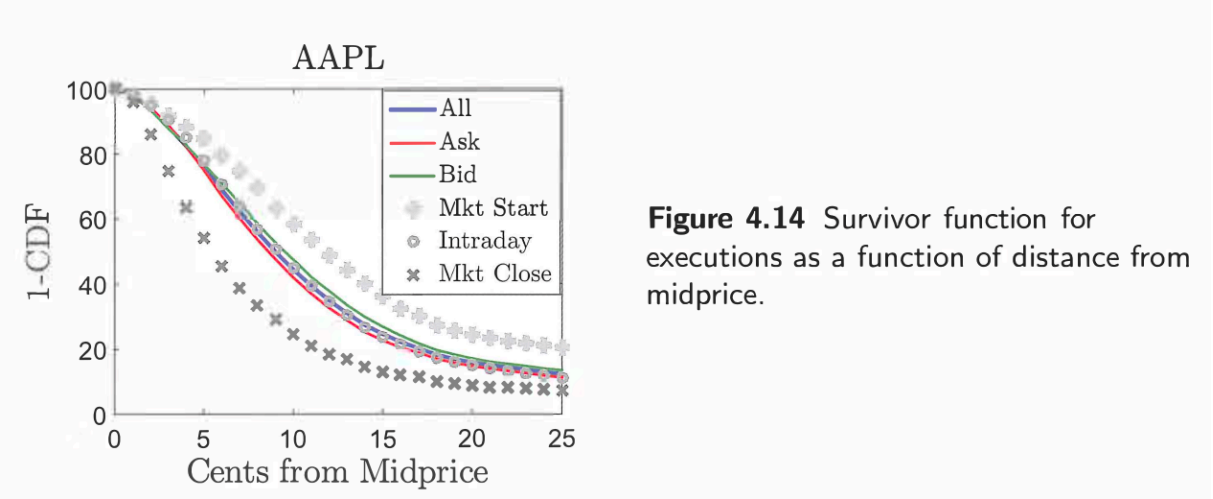


Figure 4.14 Survivor function for executions as a function of distance from midprice.

Figure 3.19: Survival function of distance from midprice.[82]

scope of this study.[84] The market share ρ will be multiplied to the cash and quantity dynamics of the market maker while the price will evolve according to all orders in the market. When he has limit orders in the bid queue, the increase in quantity will be the number of shares v of the market order hitting the bid multiplied by the market share ρ (assume market maker's limit orders are distributed evenly within the queue).[85]

This indicates that market buy orders tended to occur much closer to the midprice than market sell orders on this particular day, which had an overall positive order flow for AAPL shares and a slight price increase from market open to market close. In Figure 4.14, we have also included total executions separated by the time of day: the first half hour after the market opens (Mkt Start), the last half hour before the market closes (Mkt Close), and the time in between (Intraday). We observe that Mkt Close tends to be below that of Intraday, implying that during the last half hour of trading, executions tend to be close to the midprice, which is consistent with the pattern of the quoted spread in Figure 4.6.

3.3 Empirical Intraday activity patterns and stylized facts

As we have seen in the preceeding empirical analysis, the data exhibits the following key charactersistics, that set the stage for our benchmarking framework for the mathematical modelling and its performance assessment:[86]

1. irregular time intervals between arrivals

[84] [See Law15, p.61]
[85] [See Law15, p.62]
[86] [See Law15, p.13]

2. discrete state space of price ticks and volume lot sizes
3. intraday seasonality (more activities around market open and close)
4. arrival clustering
5. self-excitation from its own history
6. cross-excitation from other order types
7. cross-excitation from the history of other assets
8. long memory of excitation effect
9. sparse order book characteristsics (low liquidity, low market depth, wide bid-ask spread)[87]
10. Non-Markovian dynamics (the future evolution of the LOB depends not only on the current state but also on the past history of order arrivals and executions)[88]

As with the newest regulatory requirements, the todays standard will be the 15-minute contracts, especially item seven proposes a complexity increasing hurdle. Suspectively the closer a neighbouring contract is, the more cross-excitation impact will be observed, suggesting few contracts before and after the current contract, as the most relevant ones for the modelling. We chose to ignore this complexity for the sake of simplicity and focus on the single contract modelling.

[87] [See Bal22, p.236]
[88] [See CJP15, p.52]

4 Mathematical Modelling

In order to realistically model the LOB dynamics and derive the optimal market making strategy, we need to choose a suitable mathematical framework that can replicate the key empirical characteristics of the LOB. A natural framework is offered by the Hawkes-Process family, which is a class of self-exciting point processes that can capture the clustering and dependence structure of order arrivals in the LOB.[89] By also including cross-excitation among different order types, this process family also offers modelling capabilities for complex interactions.

Explicitly not assuming that the MM can adjust bid and ask prices continuously, the MM's decision problem is formulated as a form of an optimal impulse-control switching problem, see [Law15].

Not only is this assumption not realistic but also not properly penalised in the existing discussions, as altering order parameters means losing the queue position due to the applied price-time priority principle most exchanges operate on.

In the following sections, we will first introduce the basic concepts of Hawkes processes, as member of the general point process family, and then describe how we can use this framework to model the order arrivals and mid-price dynamics in the electricity markets, ultimately leading to the derivation of the optimal market making strategy. A complete taxonomy of the main process families belongig to the point processes is given below.

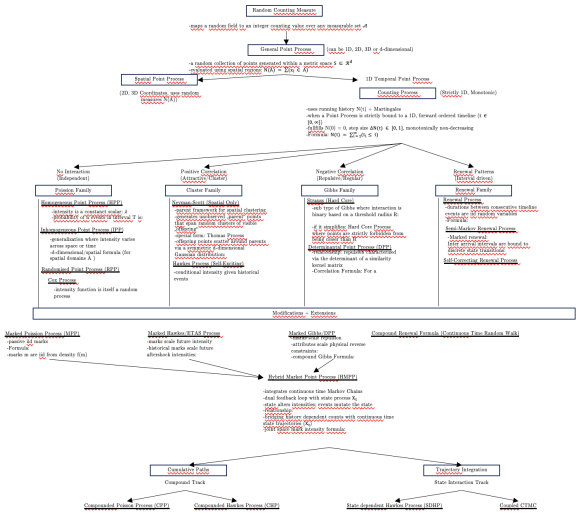


Figure 4.1: Shcematic overview of the point process family.[90]

[89] Note: In theory the order arrival intensities can be any predictable non-negative stochastic processes, [see Kor25, p.39].
[90] Based on [Kie19], own illustration.

4.1 The Point Process Family

Starting with the basic concepts of the general point process family, we can define a point process as a random collection of points in time, where each point represents the occurrence of an event. This basic model for representing event streams assumes that their occurrence is random, i.e. independent and can be described by a probability distribution.[91]

4.1.1 Homogeneous Point Processes

[See Can20, p.3]

Definition 4.1 (Point process)**.** A point process is a stochastic process $(T_k)_{k\in\mathbb{N}}$ taking values in $\mathbb{R}_+$ such that for every $k \in \mathbb{N}$, $T_k \leq T_{k+1}$.

Definition 4.2 (Counting Process)**.** Given a point process $(T_k)_{k\in\mathbb{N}}$, its counting process is defined by

$$N(t) = \sum_{k\in\mathbb{N}} \mathbf{1}_{\{T_k \leq t\}} \tag{4.1}$$

and counts the number of events that have occurred up to time t. The process is called *simple* if no two events occur at the same time, i.e., $\mathbb{P}(T_k = T_j) = 0$ for all $k \neq j$.

In reality of course, multiple incoming messages can have the exact same assigned timestamp, suggesting a *non-simple* point process. However, for the sake of simplicity, we will assume a simple point process in our modelling framework, as these cases are very rare and only make up x % of observations (timestamped on nanoseconds).[92]

An important characteristic of the counting process is its so-called *conditional intensity*:

Definition 4.3 (Conditional Intensity)**.** The conditional intensity $\lambda(t)$ of a counting process $N(t)$ is defined as the limit

$$\lambda(t) := \lim_{\Delta t \to 0} \frac{\mathbb{P}(N(t + \Delta t) - N(t) = 1 | \mathcal{F}_t)}{\Delta t} \tag{4.2}$$

where $\mathcal{F}_t$ is the filtration generated by the counting process up to time t, i.e. $\mathcal{F}_t := \sigma(N(s) : 0 \leq s \leq t)$.

Another important quantity in the study of point processes is the point-wise integral of the conditional intensity of the process, $\Lambda(t)$, i.e.

[91] [Pal43]

[92] Assume no two types of orders can arrive at the exact same time, if the exchange operates multiple servers it can of course accept two incoming order types at the same instant, [see Law15, p. 45]

Definition 4.4 (Compensator)**.**

$$\Lambda(t) := \int_0^t \lambda(s)ds \tag{4.3}$$

We call this value the compensator of the process, and we have the following useful result about compensators (Lemma X.Y.Z in XX).

Theorem 4.1 ($\mathcal{F}$-Martingale)**.** Suppose the $\mathcal{F}_t$-adapted counting process $(N(t))_{t\in[0,\infty)}$ has a left-continuous intensity $\lambda(t)$. Then the process $M(t) := N(t) - \Lambda(t)$ is an $\mathcal{F}_t$-martingale, i.e. for every $0 \leq s \leq t$,

$$\mathbb{E}[M(t)|\mathcal{F}_s] = M(s). \tag{4.4}$$

Counting processes preserve three notions of regularities, namely: *stationarity*, *simplicity* and *orderliness*.
A counting process N is said to be (strictly) *stationary* if $N(t+s) - N(s)$ has the same distribution for all $s \geq 0$. N is said to be *simple* when:

$$\mathbb{P}(N(t) = 0 \text{ or } 1 \text{ for all } t) = 1 \tag{4.5}$$

and orderly, wehn it is first stationary and:

$$\mathbb{P}(N(t+h) - N(t) \geq 2) = o(h) as\ h \to 0 \tag{4.6}$$

.

One of the most prominent members of counting processes is the *Possion Process*, that can be defined in serveral ways. Based on its intensity it may be defined as:

Definition 4.5 (Possion Process)**.** Let N be a counting process and consider a sequence $(s_i, t_i]$ of half-open intervals with $0 \leq s_i < t_i \leq s_{i+1}$, $i \in \{1, \ldots, k\}$. N is called a Poisson process with intensity $\lambda > 0$ if:

$$\mathbb{P}[N(t_i) - N(s_i) = n_i, i \in 1, \ldots, k] = \prod_{i=1}^{k} \frac{(\lambda(t_i - s_i))^{n_i}}{n_i!} e^{-\lambda(t_i - s_i)} \tag{4.7}$$

It follows from this definition that a Possion Process satisfies the following conditions:

1. The number of events in an interval $[s, t)$, $0 \leq s < t$, follows a Poisson distribution with mean $\lambda(t - s)$
2. the process has *independent* increments, i.e. the number of arrivals in disjoint intervals are independent.

Allowing the intensity λ to vary with time, we transition from the homogeneous to the non-homogeneous Possion Process.[93]

[93] [See Can20, p.4]

4.1.2 Non Homogeneous Point Processes

In a non-homogenous Poisson process, the (infinitesimal) probability of an event happening at time t may vary with t, but it is still independent of other events.[94]

An importnat characteristic of the Possion Process is its *memorylessness*, i.e. the distribution of future arrival times only depends on information available at the current time t, and not what happened in the past leading to t.[95] To adjust for this *memoryless* property, a historic dependency structure has to be introduced, leading to the *Hawkes-Process*.

4.1.3 One Dimensional Hawkes Processes

A Hawkes process[96] (Liniger, 2009) supposes that past events can temporarily raise the probability of future events, assuming that such excitation is positive, additive over the past events, and exponentially decaying with time, see [97][See ME17, p.1] Real world scenarios often violate these assumptions, as an event can also be inhibitious rather than excitatory, and the excitation effect may not be strictly additive over time as effect strengths might dampen down or that an exponentially decay can't be guaranteed.[98] The output of a Hawkes process is essentially an event intensity, which is an instantantaneous probability.[99]

Definition 4.6 (Hawkes Process)**.** A Hawkes process is a simple point process with a conditional intensity function $\lambda(t)$ defined as:

$$\lambda(t) = \mu + \int_{t_i<t} \phi(t-s)dN_s \tag{4.8}$$

where $\mu \in \mathbb{R}^+$ is the baseline intensity of the process, $\phi(t) : \mathbb{R}^+ \to \mathbb{R}^+$ is the excitation kernel function, and t_i are the times of past events.

[100]

In dimension $d = 1$, the process exhibits *self-excitation*, where past event occurences, spark an increase in the likelihood of observing events in the near future. Note that in $d = 1$, only one specific type of event is considered for the self-excitation. It is of course possible and rather plausible, that in realistsic scenarios, multiple types of events are responsible for future excitations, proposing the need to incporporate a dependency structure for mutual event type impacts.[101] To motivate a later extension, in effect, Hawkes processes and

[94] [See ME17, p.1]
[95] [See Can20, p.4]
[96] [See Haw71, p.]
[97]
[98] [See ME17, p.2]
[99] [See ME17, p.2]
[100] [See Can20, p.5]
[101] [See Can20, p.5]

continuous-time Markov chains can have either an event viewpoint or a state viewpoint, respectively.[102]

4.1.4 Multi Dimensional Hawkes Processes

[See Can20, p.5]:

Definition 4.7 (Multivariate Hawkes Process)**.** A d-dimensional counting process $N = (N_1, \ldots, N_d)$, is a multivariate Hawkes process when:

$$\lambda_i(t) = \mu_i + \sum_{j=1}^{d} \int_0^t \phi_{ij}(t-s) dN_{j,s} \tag{4.9}$$

where $\mu_i \in \mathbb{R}^d$ are the baseline intensities, t_i are the times of past events and $\phi_{ij}(t) : \mathbb{R}^+ \to \mathbb{R}^{d \times d}$ are the excitation kernel functions, and satisfy:

1. $\phi_{ij}(t) \geq 0$ for all $t \in \mathbb{R}$ and $i, j \in \{1, \ldots, d\}$
2. $\phi_{ij}(t) = 0$ for all $t \leq 0$ and $i, j \in \{1, \ldots, d\}$
3. $\phi_{ij}(t)$ is locally integrable for all $i, j \in \{1, \ldots, d\}$ i.e. $\phi_{ij}(t) \in L_{1,loc}(\Omega)$

As in XX, two alternative notations of Equation 4.9 can be considered, allowing for a more compact notation and extended interpretation of its components. The first one, being the *convolutional* notation:

$$\Lambda(t) = \mu + \Phi * dN_t \tag{4.10}$$

where $\Phi * dN_t$ is the convolution of the kernel function Φ with the counting process dN_t. The second one, being the *matrix* notation:

$$\lambda(t) = \mu + \int_0^t \Phi(t-s) dN_s \tag{4.11}$$

4.1.5 Adding State Dependencies and Marks

After having discussed the one- and d-dimensional Hawkes Process, the state dependent and marked variants should be introduced. Understanding how these two properties are essential for a realistic modelling of the LOB dynamics, we must first differentiate them along two dimensions:

1. where information originates from and
2. how it affects the intensity (λ_t) of future order arrivals.

[102] See [MP18] p.2

Adding state dependency and marks is also motivated by the findings here: [see CJP15, p.83] where we clearyl see that the size of an order (so its magnitude) impacts the arrival rates of after following orders heavily. Here it is alos suggested that order arrivals seem to be followed by further order arrivals on both sides of the book, and more on its own side than on the opposing side [see CJP15, p.83].

4.1.5.1 The Marked Hawkes Process (MHP)

The Marked Hawkes Process can be thought of as containing Event-Driven Info. The intensity changes based on a specific property attached directly to an incoming event at the exact millisecond it arrives. An incoming order has attributes like its size, direction, and price level, which are all relevant for the future order arrival intensity. The Marked Hawkes Process is therefore a natural extension of the standard Hawkes Process, where the intensity is only a function of the past event times.

4.1.5.1.1 Mathematical formulation Every time an event i occurs at time t_i, it carries a "mark" (m_i). In market microstructure, this mark is almost always the Order Size / Volume or Trade Direction:

$$\lambda_t = \mu + \sum_{t_i<t} \phi(t - t_i, m_i) \tag{4.12}$$

The intuition behind including marks is that the impact of an event on future order arrivals is not only determined by its occurrence but also by its characteristics. For example, a large market order (carrying the mark m_i) may have a more significant impact on the afterfollowing excitation ripple (a bigger shock through the kernel function: $\phi(t - t_i, m_i)$) and thus a higher intensity of future orders than a small MO, especially considering situations where the Market-depth is thin. In this thesis, we will explore if incorporating the *Volume* of an order as mark will carry any beneficial information for the future order arrival intensity.

Memorywise: The system only looks backward at the history of individual events and their specific sizes.

4.1.5.2 The State Dependent Hawkes Process (SDHP)

The State-Dependent Hawkes Process represents Environment-Driven Info. The intensity changes based on the ambient state of the entire system at time t, regardless of which specific event caused that state.

4.1.5.2.1 Mathematical formulation Introduced to fix a major flaw in pure Hawkes models: traditional Hawkes processes ignore the geometry of the order book. In a state-

dependent model, the kernel itself is modulated by a piecewise constant state variable, X_t, representing the continuous state of the LOB.

$$\lambda_t = \mu(X_t) + \sum_{t_i<t} \phi_{X_{t_i}}(t - t_i, X_t) \tag{4.13}$$

The intuition behind including state dependencies is that the state X_t could be the bid-ask spread or the queue imbalance. If the spread is at its minimum tick size, the probability of a price-improving limit order dropping in is physically zero. The model dampens or accelerates intensities based on the geometry of the book layout when the order hits.

Memorywise: The system is fully coupled. The arrival of an order changes the book state (X_t), and the new book state instantly updates the baseline (μ) and excitation physics (Φ) of future orders.

4.1.5.3 The Unified Approach: The State dependent marked Hawkes Process (SDMHP)

In a combined model, the intensity λ_t of order arrivals depends simultaneously on the changing geometry of the book (X_t) and the specific properties of past historical trades (m_i).

4.1.5.3.1 Mathematical formulation

$$\lambda_t = \mu(X_t) + \sum_{t_i<t} \phi_{X_{t_i}}(t - t_i, m_i, X_t) \tag{4.14}$$

with the following components:

- X_t: The state of the LOB at time t, which can include variables such as the bid-ask spread, queue imbalance, and other relevant market conditions.
- m_i: The mark associated with the i-th event, which can represent characteristics such as order size or trade direction.
- $\mu(X_t)$: The baseline intensity, which is a function of the current state of the LOB, capturing the underlying rate of order arrivals based on the current market conditions.
- $\phi_{X_{t_i}}(t - t_i, m_i, X_t)$: The coupled excitation kernel, which depends on the time since the last event $t - t_i$, the mark m_i, and the state of the LOB at the time of the event X_t.

This combined approach allows to properly model such cases:

> A huge trade (large mark m_i) just executed. Because the bid-ask spread is currently wide (state X_t), this massive trade will trigger a cascade of aggressive cancel-and-replace orders ($\phi_{X_{t_i}}$ becomes heavily excited). However, if the

spread were narrow, that exact same size trade would trigger virtually no reaction.

4.2 Extension: The Neural Hawkes Process (NHP)

While computationally beautiful, implementing a State-Dependent Marked Hawkes process presents a massive hurdle: Parameter Explosion. If you track a limit order book with 10 distinct states (e.g., deciles of queue imbalance) and use 3 types of marks (small, medium, and large orders), your kernel combinations multiply exponentially. Estimating these parameters in real time using Maximum Likelihood Estimation (MLE) or optimization methods like the Karush–Kuhn–Tucker (KKT) conditions becomes computationally intensive for high-frequency setups. To solve this, modern quantitative research frequently utilizes Neural Hawkes Processes (NHP). Instead of computing explicit mathematical formulas for every state-mark combination, deep learning architectures (like Continuous-Time LSTMs) digest the raw sequence of states and marks together, outputting a dynamically changing arrival intensity natively.

4.3 The Hawkes Process based framework for optimal market making

Essentially the framework shall incorporate the accurate modelling of events that impact the mid-price and the bid-ask spread, as these are the two main reference points from which we desire to obtain the optimal depths for the MM quoting policy. Further we assume that the MM quoting policy shall not be continuous, as the optimization is formulised as an optimal switching problem where the market maker will be penalized every time he revises his quotes due to the price-time priority of modern exchanges.[103] For the switching regimes, we want to obtain the best quote depths for the strategy.[104] The effective arrival rate of the market orders hitting the market maker is determined from the target market share parameter ρ chosen in advance by the market maker.[105] Volumes are used as marks and the MM is allowed to liquidate excess inventory using MO.[106] The control problem is now a combined (simultaneous) optimal switching and impulse control problem under a pure-jump environment.[107]

[103] [See Law15, p.8], this essentially assumes the MM can't reposition continuously.
[104] [see Law15, p.9]
[105] [see Law15, p.9]
[106] [see Law15, p.9]
[107] [see Law15, p.9]

4.3.1 Time-inhomogeneous (Baseline) Intensities

As we have seen in the section 3.2.6 Messages and Cancellations, the intraday activity pattern is not constant, but rather follows a certain pattern, with higher activity during certain hours of the contract lifetime, which is also known as the *Samuleson effect*. This means that the baseline intensity of the Hawkes process, which represents the underlying rate of order arrivals, has to be dependent on the remaining contract lifetime, therefore being non-stationary. By incorporating time-inhomogeneous baseline intensities into our modeling framework, we can better capture the dynamics of order arrivals and improve the performance of our market making strategy. For the baseline intensity we assume:

$$\mu_j(t) = \gamma_j e^{\eta_j t}, \quad j \in \{1, \ldots, 6\}, \tag{4.15}$$

where:[108]

Note the important practical caveat, that when the bid-ask spread is equal to the contract's tick size (δ), it is not possible to place LOs inside the bid-ask spread and the intensities of buy and sell LOs placed into the bid-ask spread should be zero.

4.3.2 Excitation Kernels

In the Hawkes-Poisson Process, the probability drop-off of an event triggered by a past event is called "kernel".[109] Together with the previously specified baseline intensities, the kernel functions determine the overall intensity of the process. The kernel functions can be generally of arbitrary functional shape, where the most common choices are the exponential, power-law, and Rayleigh kernels. A comparing figure of these kernel function types is provided below.

Here also a plot comparing the different popular choices for the exciattion kernels, but also showing the spline based non-parametric kernel estimation variant. After the empirical findings from section XX, we chose to use the exponential kernel function, specified as:

$$\phi_{j,k}(t) = \alpha_{j,k} e^{-\beta_{j,k} t}, \quad j, k \in \{1, \ldots, 6\}, \tag{4.16}$$

where:[110]

4.3.3 Midprice Dynamics

Having fully specified the order arrival rate dynamics, we can now turn to the midprice dynamics. This part represents a crucial aspect of our framework, as this often done using an arbitrary set process, that is *independent* of the order arrival events considered here,

[108] [See Luc+21, p.80]
[109] Alternatively, referred to as "de-cay/exciting/fertility kernel", [see Law15, p.19]
[110] [See Luc+21, p.80]

which is not realistic. There are two possible scenarios, either there is enough liquidity at both the best bid and the best ask levels or the opposite (a just one sided scenario will be ignored). In the first setting an incoming MO (supposedly even with bigger volume) will not cause a change in the midprice, while in the second setting even a small MO will probably cause either the bid or the ask side to be depleted, causing the midprice to change. As we have seen in the empirical section, the latter scenario is usually the default in the CID market, as the liquidity tends to be low and the bid-ask spread several tick sizes wide. Therefore we require a setting where the mid-price is impacted by our mid-price moving event types and is of the type of a pure jump process, following the considerations of [see Law15, p.7]. The directly resulting linkage between the best bid and ask prices and incoming MO (that shall be connected) can be seen here: [see Law15, p.8]

Therefore we assume it is best to not only submit orders to the best bid and ask level, but rather consider a setting where the optimal depths for posting orders should be determined, as the best bid and ask levels can be depleted.

As stated previously, we concentrate on those events that move the midprice. These are the *aggressive* events that can be depicted in Table XX and their empirical relative shares in Table XX. Namely they are: placement of buy and sell LOs into the bid-ask spread, depletion of the first or more levels of the LOB, and the placement of buy and sell MOs. [111] Therefore resulting in the use of a 6-dimensional Hawkes process. As sanity check we have to note that in the case where the bid-ask spread is just tick-size, it is of course not possible to place further LO inside the spread, so the probability of this event type is zero, and the model should be able to capture this. The mid-price must be further connected to say at least MO event types, for two reasons: First, otherwiese it can happen that a sell MO arrives and the mid-price jumps into the opposite direction if not coupled and secondly the MO carrying mark, say its attached volume must be considered. A very big MO should have a bigger impact on the mid-price as its probably gonna deplete the best few levels until its filled than say a small MO, that *does not walk the book*. Further, we will extend the proposed basis framework of [Luc+21], where his considered main mid-price moving events are the following six: placement of buy and sell LOs into the bid-ask spread, depletion of the first or more levels of the LOB, and the placement of buy and sell MOs by the following more granular extensions: bid-side MO, LO, cancellations at L1, L2, L3. ask-side MO, LO, cancellations at L1, L2, L3.

Here the mathematical definition of the midprice dynamics, which is the central variable of interest in our modeling framework. The

In reference to the before mentioned order type classification, only informed traders post aggressive order types that move the mid price, here from the data again measure the shares of these information carrying order types and show if this problem is immanent and how we can obtain the frequencies from these stats

We need to have the midprice impacted by incoming MO due to realism, cause a certain

[111] [See Luc+21, p.80]

jump of the process, jump size and direction have to be realistic (in their direction, foresign and magnitude).

Must be a (pure) jump process since it is a discrete price grid, and the midprice can only jump in discrete increments of the tick-size, which is 0.01 EUR/MWh in the CID.

As the operating frequency at which LOs are posted inside the bid-ask spread and therefore cause it and the mid-price to move is much lower when compared with traditional financial markets and occurs randomly, it is not uncommon to oberseve phases where there is no change at all.[112] A natural choice is therefore to model the mid-price and the bid-ask spread as pure jump process.[113]

Based on [See CJP15, p.262], the midprice dynamics can be used to account for adverse-selection risk a MM faces, by either assuming that the midprice undergoes jumps in the direction of incoming MO, or by letting the midprice's drift have a short term alpha component which is affected by the arrival of MO.

4.3.4 Cost functions

4.4 Overview: Modelling Specifications

Although possible in theory, the modelling of cross-excitement among various hourly products is not pursued in this thesis, as the in-parallel lifetime of the contracts differs (See Table) what makes it hard to calculate the cross-excitation, and [Zweiter Grund].

Ignoring the external influence of updated forecasts and other external variables, can be partly defended by seeing that fundamental drivers are partly not relevenat for the liquidity, see here: [HW13].

Also the idea, i.e when we refit daily, we could use these values as prior for the next fit making use of Bayesian updating, see [Jai+24a]

Here a table as overview of all different model versions that we fit and want to compare their performances in judging whether some specifications and inclusions are justified or not.

Having discussed the rich possibilities in that the modelling frameworkmcan be specified and extended, we will now turn to the specifications of the modelling frameworks we want to test:

1. We start off by the model proposed by [Luc+21], which is a 6-dimensional Hawkes process, where the midprice moving events are the following six: placement of buy and sell LOs into the bid-ask spread, depletion of the first or more levels of the LOB, and the placement of buy and sell MOs.

[112] [See Luc+21, p.78]
[113] [See Luc+21, p.78]

2. Next we separate the bid/ask event directions, considering: bid-side aggressive events, ask-side aggressive events, side-specific arrival intensities and side-specific impact on midprice and spread. This should serve us to test whether asymmetry between buy and sell flow matters.
3. On top the event types are added: market orders, limit orders, cancellations
4. Additionally we will consider not only the best bid/ask level but rather: L1, L2, L3 on bid side, L1, L2, L3 on ask side, level-specific intensities and excitation and level-specific queue depletion and replenishment.
5. Letting intensities react to the current book state is achieved by adding state dependencies using possible variables like: spread, midprice, imbalance, queue sizes at L1–L3, time-of-day / time-to-delivery, possibly volatility regime.
6. We will next add marks to incorporate event magnitudes with the following possible components: order size, price impact size, possibly depth consumed, possibly aggressiveness of the order.
7. In a more optional setting we can benchmark the so far explored settings against a Reinforcement Learning extension, with this broad formulation: Hawkes simulator, state = intensities + book state + inventory, actions = bid/ask quote placement, reward = PnL minus risk penalties.

The specifications in that order look like follows. The starting base model of [Luc+21] is a 6-dimensional Hawkes process. The vector of event counting processes $N(t)$ is given as:

$$N(t) = (N^{BM}(t)m, N^{SM}(t), N^{BL,+}(t), N^{BL,-}(t), N^{SL,+}(t), N^{SL,-}(t)) \tag{4.17}$$

where

- $N^{BM}(t)$ is the counting process for buy market orders,
- $N^{SM}(t)$ is the counting process for sell market orders,
- $N^{BL,+}(t)$ is the counting process for buy limit orders placed inside the bid-ask spread,
- $N^{BL,-}(t)$ is the counting process for buy limit orders placed outside the bid-ask spread,
- $N^{SL,+}(t)$ is the counting process for sell limit orders placed inside the bid-ask spread, and
- $N^{SL,-}(t)$ is the counting process for sell limit orders placed outside the bid-ask spread.

For each of the six processes: $i \in \{1, \dots, 6\}$ define the conditional intensities:

$$\lambda_i(t) = \mu_i(t, X_t) + \sum_{j=1}^{6} \int_0^t \phi_{ij}(t-s) dN_j(s) \tag{4.18}$$

where:

- $\lambda_i(t)$ conditional arrival intensity of event type i,
- $\mu_i(t, X_t)$ is the baseline intensity for process i, which can depend on time t and the current state of the limit order book X_t,
- $\phi_{ij}(t-s)$ is the excitation kernel function that captures the influence of from event type j to event type i,
- X_t state vector, e.g. spread, midprice, imbalance, time-to-close at time t.

The mid-price and spread dynamics can be formulated as follows. Define the mid-price as P_t and the Spread as S_t as pure jump processes, driven by the price-moving events:

$$dP_t = \sum_{i=1}^{6} \Delta_i^P dN_i(t) \tag{4.19}$$

and the spread as:

$$dS_t = \sum_{i=1}^{6} \Delta_i^S dN_i(t) \tag{4.20}$$

where

- Δ_i^P is the price impact of event type i on the mid-price,
- Δ_i^S impact of event i on the spread.

State dependency can be added by describing the state using the possible variables combined as:

$$X_t = (S_t, P_t, I_t, Q_t^{L1}, Q_t^{L2}, Q_t^{L3}, T_t, V_t) \tag{4.21}$$

where:

- S_t is the bid-ask spread at time t
- P_t is the mid-price at time t
- I_t is the order imbalance at time t
- $Q_t^{L1}, Q_t^{L2}, Q_t^{L3}$ are the queue sizes at levels 1, 2, and 3 of the limit order book at time t
- T_t is the time-to-delivery at time t
- V_t is the volatility regime at time t

Then a baseline can be chosen as:

$$\mu_i(t, X_t) = f_i(S_t, P_t, I_t, Q_t^{L1}, Q_t^{L2}, Q_t^{L3}, T_t, V_t) \tag{4.22}$$

or alternatively:

$$\mu_i(t, X_t) = exp(\alpha_i + \beta_i^\top X_t + g_i(t)) \tag{4.23}$$

where:

- α_i channel-specific level,

- β_i sensitivity to the state variables,
- $g_i(t)$ deterministic intraday seasonality..

Including marks m_k to each event, for example the order size as mark, a marked kernel can be written as:

$$\phi_{ij}(u, m) = a_{i,j} h(m) e^{-\omega_{i,j} u} \mathbb{1}_{u>0} \tag{4.24}$$

where:

- $a_{i,j}$ strength of excitation,
- $\omega_{i,j}$ decay speed,
- $h(m)$ mark impact function, e.g. $h(m) = m$ or $h(m) = \log(1 + m)$

what lets larger MOs or LOs have bigger effects than smaller ones.

The level resolved extension results from refining each aggregated event type further by side, event type and level, resulting in a 18-dimensional Hawkes process: $i \in \{$side x type x level$\}$ with:

- side $\in \{$bid, ask$\}$,
- type $\in \{$MO, LO, CA$\}$,
- level $\in \{1, 2, 3\}$.

with the intensities becoming:

$$\lambda_i(t) = \mu_i(t, X_t) + \sum_{j=1}^{18} \int_0^t \phi_{ij}(t - s, m) dN_j(s), i = 1, \ldots, 18 \tag{4.25}$$

Stability conditions: For exponential kernels, define the integrated excitation matrix G as:

$$G_{ij} = \int_0^\infty \phi_{ij}(u) du = \frac{a_{i,j}}{\omega_{i,j}} \tag{4.26}$$

A sufficient stability condition is $\rho(G) < 1$ with:

- $\rho(G)$ is the spectral radius of the matrix G,
- G_{ij} is the expected number of events of type i triggered by a single event of type j.

4.5 Deriving the optimal strategy: the optimal depths

Here the derivation and solution of the stochastic optimal control problem for the optimal depths, which is

4.5.1 Stochastic Optimal Control

The goal of stochastic control problems is to maximise (minimise) some expected profit (cost) function by tuning a strategy which itself affects the dynamics of the underlying stochastic system and to find the strategy which attains the maximum (minimum).[114]

A prominent tool for solving such problems is the *dynamic programming principle* (DPP) and the related non-linear partial differential equation (PDE), also knwon as *Hamilton-Jacobi-Bellman* (HJB) equation.[115] With the DPP a stochastic control problem can be solved by breaking it down into a sequence of smaller subproblems, which can be solved recursively. The HJB equation provides a necessary condition for optimality and can be used to derive the optimal control strategy by solving the PDE subject to appropriate boundary conditions.[116]

In the context of this thesis, we are especially interested in the setting where a MM wishes to maximise expected terminal wealth by using MO and LO.

4.5.1.1 Control for Diffusion Processes

4.5.1.2 Control for Counting Processes

4.6 Estimation techniques

Here describe the maximum likelihood or GMM estimations for the various model parameters, before we head over to the next pratical chapter where we apply all together.

[114] [See CJP15, p.100]
[115] [See CJP15, p.100]
[116] [See CJP15, p.100]

5 Numerical Analysis and Simulation

Here the concrete estimation procedures for the arrival rates and intensities

5.1 Modelling Order arrivals

Since the LOB and every state can essentially be reconstructed by the two main input parameters, that are the arrival of limit orders and market orders, we can focus on the modelling of these two main order types, as well as cancellations, which are the main drivers of the LOB dynamics. From these general order types, we further focus on those that are actually relevant, i.e. those that move the mid price, which are the so called *aggressive orders*, according to the detailed order-categorization based on XYZ laid out earlier in chapter XYZ. Another central and shaping input is the volume depth of the LOB.[Cho+21] The basic idea behind modelling order arrivals is the following one: some event types are stochastically excited or inhibited—in the sense of having their probabilities elevated or decreased—by patterns in the sequence of previous events.[117] Learning this pattern enables us to predict which type of event will happen next and when.[118]

Remembering the critical consideration, that a MM has no intrinsic need or desire to hold any inventory, so she will only buy (sell) in anticipation of a subsequent sale (purchase), what motivates the pursued modelling of order arrivals.[119]

The estimated relative shares per order type here from the data:

Table 4.2.
Summary statistics of QQQ on June 2, 2014 (12pm-2pm)

Type	Description	Count	% Count	Avg Rate (/s)	Avg Size
1	aggressive market buy	190	0.08%	0.0264	1,040
2	aggressive market sell	171	0.08%	0.0238	667
3	aggressive limit buy cancellation	164	0.07%	0.0228	685
4	aggressive limit sell cancellation	106	0.05%	0.0147	544
5	non-aggressive market buy	312	0.14%	0.0433	670
6	non-aggressive market sell	210	0.09%	0.0292	874
7	non-aggressive limit buy	54,187	24.19%	7.5260	1,006
8	non-aggressive limit sell	53,332	23.80%	7.4072	1,061
9	non-aggressive limit buy cancellation	58,481	26.10%	8.1224	897
10	non-aggressive limit sell cancellation	56,265	25.11%	7.8146	927
11	aggressive limit buy	316	0.14%	0.0439	842
12	aggressive limit sell	315	0.14%	0.0438	681

Figure 4.2 shows the summary statistics of Powershares QQQ Trust (Nasdaq 100 index ETF) (QQQ). It is well-known that the activities of limit order revisions dominate the market and it is reflected clearly in the table that 99.2% of the orders are type 7-10.

Figure 5.1: Relative Percentage share of each order type.[120]

[117] [See ME17, p.1]
[118] [See ME17, p.1]
[119] [See CJP15, p.21]
[120] Based on [Law15], own illustration.

5.1.1 Market Order Arrivals

5.1.2 Limit Order Arrivals

5.1.3 Cancelations

5.1.4 The role of Order Imbalance

5.2 Exchange Latency

Here the exact calculation of the exchange latency is provided, which is the time it takes for an order to be processed and executed on the exchange after it has been submitted. This includes the time taken for the order to reach the exchange, be processed by the exchange's matching engine, and receive a confirmation of execution. The exchange latency can vary based on several factors, including the type of order, the current market conditions, and the specific exchange being used. Understanding and minimizing exchange latency is crucial for developing effective trading strategies, particularly in high-frequency trading environments where even small delays can impact profitability. important for sampling and fitting durations etc.

5.3 Goodness of Fit Evaluation

QQ Plots here, AIC table, etc.

5.4 Optimal Strategy Backtests

Here some example graph that is shown in the List of Figures:

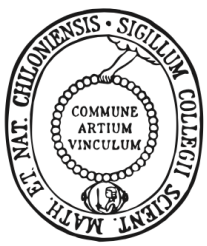


Figure 5.2: Example Figure.[121]

Here some example table that will be shown in the List of Tables section: listing example

[121] Test cite [LV19], as of 16.05.2026.
[122] See first: [Deu23].



Figure 5.3: Some text.[122]

Column 1	Column 2	Column 3
Data 1	Data 2	Data 3
Data 4	Data 5	Data 6

Table 5.1: Example Table

that will be shown in the List of Algorithms section:

Listing 5.1: Example Algorithm

```
def example_algorithm(data):
    # Some example algorithm code
    result = data * 2
    return result
```

Her some example algorithm that will appear in the List of Algorithms section:

Algorithm 5.1 Example Algorithm

Require: *data* - Input data for the algorithm
Ensure: *result* - Output result of the algorithm
1: Initialize *result* to 0
2: **for** each element in *data* **do**
3: Update *result* based on the element
4: **end for**
5: **return** *result*

Here some examples of different citation styles offered by LaTeX: Antonio J and Luis A

[AL22]. Parenthetical citation: [AL22]. Footnote citation: [AL22]. Footnote citation with page number: [AL22, p. 1]. Inline citation: Rosenberg [Ros03]. Inline citation: Antonio J and Luis A [AL22]. Inline citation: Seabold and Perktold [SP10]. Inline citation: Seabold and Perktold [SP10]. Inline citation: Fama and French [FF16]. Inline citation: Rosenberg [Ros03]. Inline citation: Fama and French [FF16]. Example of a footnote citation.[123].

An example of an equation (numbered and centered) is shown below:

$$E = mc^2 \tag{5.1}$$

[123] [FF16]

6 Results

Here the results text.

7 Discussion

Here the discussion text.

8 Conclusion

Here the conclusion text.

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A Supplementary Market Structure Information

A.1 Hourly Products and Trading Durations

Hourly Product	Delivery Interval (CET)	Cross-Border Gate Closure	Max Cross-Border Trading Duration	Local German Gate Closure	Max Local German Trading Duration	Next-Day Hours Open in Parallel	Duration of This Parallel Window
Hour 01	00:00 – 01:00	23:30 (D-1)	8 h 30 min	23:55 (D-1)	8 h 55 min	24	8 h 55 min
Hour 02	01:00 – 02:00	00:30 (D)	9 h 30 min	00:55 (D)	9 h 55 min	23	1 h 00 min
Hour 03	02:00 – 03:00	01:30 (D)	10 h 30 min	01:55 (D)	10 h 55 min	22	1 h 00 min
Hour 04	03:00 – 04:00	02:30 (D)	11 h 30 min	02:55 (D)	11 h 55 min	21	1 h 00 min
Hour 05	04:00 – 05:00	03:30 (D)	12 h 30 min	03:55 (D)	12 h 55 min	20	1 h 00 min
Hour 06	05:00 – 06:00	04:30 (D)	13 h 30 min	04:55 (D)	13 h 55 min	19	1 h 00 min
Hour 07	06:00 – 07:00	05:30 (D)	14 h 30 min	05:55 (D)	14 h 55 min	18	1 h 00 min
Hour 08	07:00 – 08:00	06:30 (D)	15 h 30 min	06:55 (D)	15 h 55 min	17	1 h 00 min
Hour 09	08:00 – 09:00	07:30 (D)	16 h 30 min	07:55 (D)	16 h 55 min	16	1 h 00 min
Hour 10	09:00 – 10:00	08:30 (D)	17 h 30 min	08:55 (D)	17 h 55 min	15	1 h 00 min
Hour 11	10:00 – 11:00	09:30 (D)	18 h 30 min	09:55 (D)	18 h 55 min	14	1 h 00 min
Hour 12	11:00 – 12:00	10:30 (D)	19 h 30 min	10:55 (D)	19 h 55 min	13	1 h 00 min
Hour 13	12:00 – 13:00	11:30 (D)	20 h 30 min	11:55 (D)	20 h 55 min	12	1 h 00 min
Hour 14	13:00 – 14:00	12:30 (D)	21 h 30 min	12:55 (D)	21 h 55 min	11	1 h 00 min
Hour 15	14:00 – 15:00	13:30 (D)	22 h 30 min	13:55 (D)	22 h 55 min	10	1 h 00 min
Hour 16	15:00 – 16:00	14:30 (D)	23 h 30 min	14:55 (D)	23 h 55 min	9	1 h 00 min
Hour 17	16:00 – 17:00	15:30 (D)	24 h 30 min	15:55 (D)	24 h 55 min	8	1 h 00 min
Hour 22	21:00 – 22:00	20:30 (D)	29 h 30 min	20:55 (D)	29 h 55 min	3	1 h 00 min
Hour 23	22:00 – 23:00	21:30 (D)	30 h 30 min	21:55 (D)	30 h 55 min	2	1 h 00 min
Hour 24	23:00 – 24:00	22:30 (D)	31 h 30 min	22:55 (D)	31 h 55 min	1	1 h 00 min

Note: All 24 next-day hours open together at 15:00 CET (D-1). The count decreases as each individual product hits its respective local gate closure.

Table A.1: EPEX Spot CID Trading Durations and Parallel Product Counts.[124]

[124] Based on [Kie19], as of 16.05.2026.

B Data

B.1 LOB Data Description

Column Name	Description
Timestamp	The date and time of the order event
Order Type	The type of order (e.g., limit order, market order, cancellation)
Price	The price at which the order was placed or executed
Quantity	The quantity of the order
Side	The side of the order (buy or sell)
Order ID	A unique identifier for the order
Trader ID	A unique identifier for the trader who placed the order
Venue	The trading venue where the order was placed
Status	The status of the order (e.g., filled, partially filled, canceled)
Time in Force	The duration for which the order is valid (e.g., GTC, IOC)
Execution ID	A unique identifier for the execution event (if applicable)
Counterparty ID	A unique identifier for the counterparty involved in the trade (if applicable)
Order Book State	A snapshot of the limit order book at the time of the event (optional)
Market Conditions	Additional market conditions at the time of the event (optional)
Notes	Any additional notes or comments related to the order event (optional)

Table B.1: Data Columns and Their Descriptions.[125]

Here the first part of the appendix text.

B.2 Second Part

Here the second part of the appendix text.

[125] Based on the typical structure of order book data, as of 16.05.2026.

C Appendix B

D Appendix C

E Code

Here some coding snippets:

Listing E.1: Example Python Code

```
    def example_function(x):
        return x**2

```

F Engineering Configurations

F.1 Hyperparameter Optimization

F.2 Computation Time

Affirmation

General Declaration of Authorship

I hereby declare that I have composed my Master's thesis *"Optimal Market Making in Intraday Electricity Markets via Hawkes Process Modeling"* independently using only those resources mentioned, and that I have as such identified all passages which I have taken from publications verbatim or in substance. I agree that the work will be reviewed using plagiarism testing software. Neither this paper, nor any extract of it, has been previously submitted to an examining authority, in this or a similar form. I have ensured that the written version of this thesis is identical to the version saved on the enclosed storage medium.

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